Journal of Industrial Information Integration 54 (2026) 101224

Contents lists available at ScienceDirect

Journal of Industrial Information Integration

journal homepage: www.sciencedirect.com/journal/journal-of-industrial-information-integration

Full Length Article

A Q-learning based hybrid heuristic algorithm for multi-truck and multi-drone routing problem in post-disaster relief distribution

Xiaofei Kou [a], Tianlong Liu [a], Hao Liu [a], Gangyan Xu [b], Peng Sun [c,*]

[a] School of Management Engineering, Zhengzhou University of Aeronautics, Zhengzhou, 450046, China
[b] Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hong Kong, China
[c] College of Management and Economics, Tianjin University, Tianjin, 300072, China

ARTICLE INFO

Keywords:
Multi-truck and multi-drone system
Vehicle routing problem
Q-learning based hybrid heuristic
Post-disaster relief distribution

ABSTRACT

The truck-drone cooperative delivery system has gained widespread attention in post-disaster relief distribution due to its flexibility, rapid response capability, and ability to operate in complex terrain. Moreover, cooperative truck–drone delivery remains highly complex, while most existing studies either keep a single truck, restrict each truck to a single drone, or relax delivery deadlines to soft penalties. In contrast, this paper introduces a vehicle routing problem for a collaborative relief distribution system with flexible coordination modes and urgency-aware delivery deadlines. A mixed integer programming (MIP) model is constructed for a multi-truck multi-drone collaborative relief distribution system (VRPMTMD) with the objective of minimizing cost, in which the delivery deadline of each site is contracted according to its urgency priority. To solve this problem, a Q-learning-based hybrid heuristic (QL-IPSO-ALNS) algorithm is proposed, which incorporates the global search capability of improved particle swarm optimization (IPSO), the local refinement ability of adaptive large neighborhood search (ALNS) and an adaptive operator selection mechanism using Q-learning to improve solution quality and computational efficiency. We verify the effectiveness of the QL-IPSO-ALNS algorithm using modified Solomon instances and a real-world case study. Experimental results show that the proposed algorithm produces solutions consistent with Gurobi on small- and medium-scale instances, while outperforming IPSO, K-means-ALNS, LNS, CH-SaBO, and MAAGLNS on large-scale instances. The findings further indicate that coordinated truck–drone deployment yields significant cost reductions. Sensitivity analyses reveal that fleet composition, urgency-differentiated deadlines, drone endurance, and objective weighting all play key roles in achieving these savings.

1. Introduction

In recent years, the global incidence of extreme meteorological disasters has intensified, posing serious threats to socioeconomic stability and public safety. Events such as typhoons, floods, and torrential rainfall have become increasingly frequent and destructive. For example, on July 20, 2021, Zhengzhou, China, experienced a catastrophic rainstorm that caused 302 deaths, including 14 fatalities in a subway tunnel and six in a highway tunnel [1]. In July 2023, Beijing was hit by an unprecedented rainfall event, with average cumulative precipitation exceeding 330 mm and breaking historical records dating back to 1961. Among the seven affected districts were Fangshan, Mentougou, and Changping, severely affected, and the maximum rainfall recorded at a single station exceeded 1000 mm. This event caused widespread damage to critical infrastructure, including power systems, communication networks, water conservancy facilities, and transportation networks [2]. Following such extreme meteorological disasters, affected areas often face a sharp increase in urgent relief demands within a very short period, with multiple affected sites requiring timely and coordinated assistance. The large number and spatial distribution of these sites, together with strict time requirements, place considerable pressure on relief distribution. At the same time, damaged or disrupted transportation networks may further increase the difficulty of reaching some affected sites and organizing efficient deliveries. Therefore, efficiently distributing relief supplies to multiple affected locations while meeting stringent time requirements and adapting to changing transportation conditions has become a critical challenge in disaster response.

Drones have attracted growing scholarly attention in logistics and

* Corresponding author.
E-mail addresses: kxfei1987@zua.edu.cn (X. Kou), tianlong@zua.edu.cn (T. Liu), lhr3282@zua.edu.cn (H. Liu), gagexgy@gmail.com (G. Xu), sunpeng1317@tju.edu.cn (P. Sun).

https://doi.org/10.1016/j.jii.2026.101224

Available online 5 October 2026
2452-414X/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

transportation research because of their high maneuverability, ease of deployment, and rapid response capability. As an emerging aerial delivery mode, drones are less dependent on ground infrastructure and can provide fast and flexible delivery services, particularly for time-sensitive and hard-to-access deliveries. In 2013, Amazon launched its Prime Air drone delivery initiative and conducted a series of pilot tests to examine the feasibility of using drones for point-to-point parcel delivery and transportation between warehouses and sorting centers. Subsequent trials in 2016 showed that the system could deliver packages weighing up to five pounds within approximately 30 min [3]. In Iceland, the e-commerce company AHA partnered with Flytrex to deploy drone delivery services in Reykjavík, integrating drones into existing delivery networks and significantly reducing parcel delivery times [4]. In 2019, Wing, a drone company under Alphabet, launched a public drone delivery trial in Canberra, Australia, initially serving around 100 households with plans for phased expansion [5].

Although drones generally travel faster and can follow more flexible routes with less reliance on road networks, their operations remain constrained by limited payload capacity, battery endurance, and airspace management regulations. These limitations reduce their practicality as a standalone delivery mode [6]. To overcome such constraints, collaborative truck-drone operations have emerged as an important research direction. In this mode, trucks serve not only as transport vehicles but also as mobile launch and recovery platforms for drones, which can be dispatched to serve affected sites and then rendezvous with the truck [6]. To improve service quality and reduce transportation cost, Ha et al. [7] investigated a variant of the traveling salesman problem with drone (TSP-D) that minimizes both transportation cost and time lost due to vehicle waiting. Agatz et al. [8] developed exact integer programming models and a set of heuristic algorithms for the synchronized routing of trucks and drones. The practical feasibility of truck-drone collaboration has also been demonstrated in industrial applications. A 2017 UPS experiment involved a roof-mounted drone that autonomously delivered a package to a residence and then rejoined the moving delivery vehicle, while the driver continued driving. Mercedes-Benz presented a concept delivery van at CES equipped with an integrated drone nest on its roof and a mobile warehouse interior [9]. In 2018, Amazon filed a patent proposing the use of third-party commercial trucks as temporary landing platforms for drones, enabling drones to “hitchhike” on trucks to conserve battery power [10]. As this operational mode continues to mature, its application is expanding from commercial logistics to broader public-service contexts.

Despite these advances, emergency logistics differ fundamentally from conventional commercial logistics, as relief operations must simultaneously satisfy stringent requirements for timeliness, service coverage, and transportation capacity in post-disaster environments. After a disaster, affected sites are often geographically dispersed, demand levels vary substantially across locations, and road accessibility may change rapidly because of flooding, debris, congestion, or infrastructure damage. Under such conditions, ground-only delivery may suffer from long detours, delayed service, and incomplete coverage. Truck-drone collaboration therefore provides a promising operational mode for emergency relief distribution. Trucks can transport relatively large quantities of relief supplies and serve as mobile launch and recovery platforms, while drones can bypass damaged or congested road segments and provide rapid last-mile delivery to remote, urgent, or difficult-to-access affected sites. The combination of ground transportation capacity and aerial mobility is therefore particularly valuable in disaster environments where both rapid response and broad spatial coverage are required.

Nevertheless, a single truck-drone system has limited service capacity, spatial coverage, and ability to handle multiple urgent tasks simultaneously. Such a configuration is insufficient for large-scale emergency distribution involving numerous affected sites and limited response times. This limitation motivates the coordinated deployment of multiple trucks and multiple drones. In a multi-truck multi-drone system, trucks undertake high-capacity transportation and provide mobile operating platforms, while multiple drones can perform short-range delivery tasks in parallel. Compared with a single truck-drone unit, this configuration can expand service coverage, increase delivery parallelism, reduce response delays, and improve the utilization of limited ground and aerial resources. It is therefore particularly suitable for post-disaster relief scenarios in which a large number of geographically dispersed affected sites must be served rapidly. However, the increased number of trucks and drones also introduces substantial coordination complexity, requiring vehicle routing, task assignment, drone operations, and temporal synchronization to be jointly considered.

Specifically, several challenges arise in multi-truck multi-drone emergency delivery. First, affected sites must be assigned not only to different trucks but also to appropriate service modes, namely direct truck service or drone-assisted service. An inappropriate assignment may lead to inefficient truck routes or excessive drone utilization. Second, for each drone mission, a feasible launch node, service node, and recovery node must be determined along the route of its carrying truck while satisfying payload and endurance constraints. Third, multiple drones carried by the same truck may conduct missions in parallel, which introduces resource competition and sequencing conflicts when several tasks require overlapping operational periods. Fourth, truck and drone operations must remain temporally synchronized. A truck may need to wait for a returning drone before continuing its route, and such waiting can propagate through subsequent delivery operations and affect the overall completion time. Finally, emergency relief tasks are associated with heterogeneous urgency levels and delivery deadlines. Although delivery deadlines specify when a task should be completed, they do not by themselves indicate the relative priority of different affected sites. This distinction is important in emergency operations because limited fast-response resources, such as drones, need to be prioritized for more time-critical tasks. Therefore, service sequencing and resource allocation need to jointly consider travel time, service duration, waiting time, and urgency-related time requirements. These interdependent decisions substantially increase the complexity of the problem and distinguish multi-truck multi-drone emergency delivery from conventional vehicle routing and single-truck drone routing problems.

To address these coordination challenges, this paper proposes a flexible collaborative delivery model for the multi-truck multi-drone emergency relief distribution problem. In the proposed system, each truck is equipped with multiple drones, and drones can be launched and recovered at visited nodes along truck routes, allowing multiple drone missions to be performed flexibly and in parallel. The model explicitly incorporates truck routing, service-mode selection, drone task assignment, launch and recovery decisions, truck-drone synchronization, waiting time, service duration, and drone payload and endurance limitations. Each affected site is associated with a time requirement determined by service availability and delivery deadline, while the deadline is adaptively adjusted according to the assessed urgency of the affected site. Thus, urgency is not introduced as an additional time constraint; rather, differences in emergency priority are translated into differentiated delivery deadlines, which help prioritize fast-response resources such as drones for more time-critical tasks. These urgency-aware requirements are incorporated into task assignment, delivery sequencing, and truck-drone synchronization decisions. Within this framework, trucks and drones are jointly scheduled to serve all affected sites while satisfying operational constraints and minimizing the overall delivery objective, which jointly accounts for transportation cost and delivery time. The proposed model thus provides a unified framework for coordinating heterogeneous ground and aerial resources in large-scale post-disaster relief distribution.

The main contributions of this study are as follows:

(1) This study develops a mixed-integer optimization formulation for an emergency logistics routing problem involving a coordinated fleet of trucks and drones operating under time-window requirements. Existing truck-drone routing studies, such as Murray and Chu [6] and Kitjacharoenchai et al. [11], consider only a subset of these operational characteristics. In contrast, the proposed framework simultaneously incorporates truck-drone cooperation, flexible drone launch and recovery locations, truck-drone interaction mechanisms, and time-window constraints within a unified modeling framework. More importantly, the formulation explicitly captures the operational dependencies between trucks and drones and represents the concurrent execution of drone missions. By integrating route planning, synchronization requirements, payload limitations, flight-endurance constraints, service time, and urgency-aware delivery deadlines, the model provides a more realistic representation of large-scale disaster-response operations. In particular, urgency is not introduced as an additional time constraint; rather, differences in emergency priority are translated into differentiated delivery deadlines, which guide the allocation of scarce fast-response resources such as drones.

(2) To solve the model proposed in this paper, a Q-learning-based hybrid heuristic (QL-IPSO-ALNS) algorithm is proposed, integrating the global search capability of improved particle swarm optimization (IPSO), the local refinement ability of adaptive large neighborhood search (ALNS), and Q-learning-based adaptive operator selection. A three-layer encoding and decoding scheme is developed to represent the decision variables and facilitate the PSO-based search, together with population initialization and particle stratification strategies to improve the quality of initial solutions. Targeted destroy and repair operators are introduced into the ALNS framework, with Q-learning employed to adaptively select operators and enhance search performance.

(3) Extensive computational experiments are conducted on benchmark datasets derived from modified Solomon instances. The proposed approach is evaluated by comparing it with Gurobi on small- and medium-scale instances and with IPSO, K-means-ALNS, LNS, CH-SaBO, and MAAGLNS on large-scale instances. The results demonstrate that QL-IPSO-ALNS algorithm consistently achieves superior solution quality. Based on the designed instances, this paper further conducts sensitivity experiments from the perspectives of fleet configuration, spatial distribution of demand, drone endurance, and objective weight allocation. Based on the computational and sensitivity analysis results, several managerial implications for relief distribution can be derived.

The remainder of this paper is organized as follows: Section 2 reviews existing studies related to the problem. Section 3 formulates the VRPMTMD and presents the mathematical model. Section 4 introduces the QL-IPSO-ALNS algorithm developed to solve the problem. Section 5 presents numerical experiments and discusses the associated sensitivity analyses. Section 6 provides a real-world case to verify the effectiveness of the model and algorithm. Section 7 draws managerial insights from the computational findings. Section 8 concludes the study and outlines several promising research directions.

2. Literature review

This study investigates the multi-truck, multi-drone collaborative delivery problem in emergency supplies dispatch and proposes a hybrid meta-heuristic algorithm for its solution. This section reviews the related literature from two perspectives, namely emergency supplies dispatch and truck-drone routing problems, and further summarizes the main solution methodologies in this field.

2.1. Truck-drone collaborative routing problem

The collaborative delivery of trucks and drones was first proposed by Murray and Chu [6] and has subsequently received great attention. Murray and Chu [6] introduced the Flying Sidekick Traveling Salesman Problem (FSTSP), in which a truck carries a drone that launches from the truck at a customer node or the depot, delivers one parcel, and is recovered by the truck. Agatz et al. [8] reformulated this setting as the traveling salesman problem with drone (TSP-D), proved approximation bounds, and solved instances with up to 100 customers using a route-first cluster-second heuristic. Yoon [12] introduced a variant of the FSTSP in which a truck can deploy multiple drones and retrieve them, tested on instances with up to 10 customers. Murray and Raj [13] further allowed heterogeneous drones with different speeds and payloads, explicitly modeling launch/recovery queueing and service times, and solved instances with up to 100 customers in about 17 min. Poikonen and Golden [14] developed the multi-visit drone routing problem (k-MVDRP), in which each drone flight can serve several customers under weight-dependent energy consumption, reporting a 58% makespan reduction when the number of drones increased from one to eight.

The above variants are confined to a single truck, whereas post-disaster relief distribution involves multiple trucks serving scattered affected sites; we therefore pay more attention to the vehicle routing problem with drones (VRPD), a generalization of the FSTSP first proposed by Wang et al. [15]. In the VRPD, packages are delivered by multiple trucks and drones, and each truck can carry a certain number of drones, which can be launched and retrieved at the depot or customer nodes. Wang et al. [15] derived worst-case bounds on the time savings achievable by introducing drones, and Poikonen et al. [16] strengthened its theoretical foundation through extended formulations and a combined time-and-cost objective. Based on Wang et al. [15], Schermer et al. [17] allowed vehicles to be equipped with multiple drones and to wait for drones during drone delivery, establishing a MILP model including drone allocation and scheduling together with a set of effective inequalities. Kitjacharoenchai et al. [18] extended the FSTSP to multiple trucks and drones with cross-truck recovery, solving TSPLIB instances with up to 99 nodes. The closest precursor to the present study is the two-echelon VRPD of Kitjacharoenchai et al. [11], in which each truck carries multiple drones that operate in parallel, serve several customers per flight, and return to their origin truck for battery swap; solved by a construction heuristic and a large neighborhood search, this model attained average savings of 13.25% over truck-only routing on instances with up to 75 customers.

Exact optimization methods have been widely used to establish optimal solutions for truck–drone routing problems. Branch-and-bound approaches [19,20] and branch-and-cut methods [21] can provide optimality guarantees, but their computational burden increases rapidly with problem size. This has encouraged the use of metaheuristic methods, which offer greater scalability for medium- and large-scale instances. Ha et al. [7] proposed a GRASP-based method for TSP-D with an effective tour-splitting procedure, while Wang et al. [22] developed INSGA-II for the bi-objective TSP-D by integrating label-based decoding with post-optimization to obtain high-quality Pareto solutions. Huang et al. [23] applied ant colony optimization to VRP-D and reported significant cost reductions compared with classical VRP solutions. For multi-truck multi-drone settings, Lan and Suzuki [24] introduced a simulated annealing-based method with intermediate launch and recovery points, improving operational flexibility and delivery efficiency. Luo et al. [25] developed an improved artificial bee colony algorithm integrating tabu search and simulated annealing, achieving better solution quality and stability for heterogeneous multi-drone systems. Xiao et al. [26] developed an improved adaptive large neighborhood search (IALNS) algorithm incorporating several novel operators tailored to the characteristics of steep roads, achieving superior performance in minimizing total energy consumption for cooperative truck-drone rural last-mile delivery systems.

More recently, learning mechanisms have been incorporated into operator selection for vehicle routing problems. Guo, T., et al. [27] proposed the learning-aided neighborhood search (LaNS), in which two reinforcement learning agents with deep Q-networks and an image-like tensor encoding of route structure adaptively select perturbation and improvement operators within a variable neighborhood search framework, achieving near state-of-the-art results on capacitated, multi-depot, and disaster-flavored VRPs. Shi et al. [28] integrated Q-learning into an NSGA-II framework to adaptively select neighborhood operators for dynamic truck–drone emergency delivery under secondary disasters. Beyond operator selection, Guo, T., et al. [29] developed a multitree genetic programming with behavioral semantics (MrGPBS) that guides the evolutionary process itself by preserving behaviorally compatible substructures, demonstrating improved search efficiency for large-scale dynamic air traffic flow management where routing and sequencing decisions are strongly coupled.

2.2. Emergency supplies dispatch problem

Road transportation has underpinned nearly all disaster operations across the preparedness, response, and recovery phases [30], and trucks remain the primary carriers of relief distribution owing to their adaptability to damaged road networks and their large payloads. Al Theeb and Murray [31] integrated relief distribution, casualty evacuation, and personnel transport into a single mixed-integer program solved by a multi-stage iterative heuristic, motivated by a refugee-camp context. Shi et al. [32] sought to maximize the total weight of transported supplies through a vehicle substitution mechanism, and Zhang and Liu [33] developed a multi-period online optimization scheme for scheduling under missing transportation-time information. These studies focus primarily on truck-based relief distribution and do not consider the complementary role of aerial delivery, which can provide greater flexibility and responsiveness when serving dispersed affected sites.

This limitation motivated the introduction of drones into humanitarian routing. Lu et al. [34] formulated a multi-objective humanitarian pickup-and-delivery routing problem in which each truck carries one drone and drone-only demand points inaccessible to trucks are served during drone flights launched from truck stops, minimizing the makespan while maximizing minimum demand satisfaction; a Wuhan COVID-19 case demonstrated the superiority of the collaborative mode over pure truck and drone systems. Zhang et al. [35] studied synchronized truck–drone delivery with site priorities, showing through a Wenchuan-earthquake case that drone speed and endurance require careful calibration. These early works established the operational value of truck–drone synchronization in relief distribution, but adopted restrictive fleet configurations – one drone per truck [34] or a single truck–drone pair [35] – and replaced delivery deadlines with priority ordering rather than enforcing them as constraints.

Subsequent research expanded the coordination scope toward multimodal and multi-depot settings. Long et al. [36] developed a dynamic truck–UAV binding strategy in which drones carried by multiple trucks may land on any truck, and solved the integrated scheduling model with a tailored tabu search, demonstrating improved resilience under road disruptions. Weng et al. [37] extended truck–drone routing to multi-depot emergency logistics and proposed a K-means-based hybrid heuristic, validated on Solomon-derived instances and a real Chengdu case. Shi et al. [38] integrated helicopters, trucks, and drones in a unified framework for areas reachable only by air, and Guo et al. [39] introduced dedicated dispatch vehicles that transport drones without forcing trucks to wait. These studies substantially enrich the coordination modes available to planners; yet each relaxes some dimension of temporal rigor – service deadlines are either absent [27,37], substituted by aggregate time objectives [38], or imposed as soft lateness penalties [36] – and fixed per-site service times are rarely modeled together with synchronization waits.

A parallel stream addresses uncertainty in post-disaster logistics. Yin et al. [40] developed a robust formulation under uncertain demands and truck travel times with delivery deadlines, solved by an exact branch-and-price-and-cut algorithm; Van Steenbergen et al. [41] applied deep reinforcement learning to dynamic scheduling under stochastic travel times; and Shi et al. [28] modeled dynamically evolving demand caused by secondary disasters, combining Q-learning with NSGA-II. At the tactical allocation level, Sun et al. [42] incorporated priority zoning and fairness considerations into truck–drone relief allocation. These contributions greatly improve robustness and responsiveness; however, because uncertainty dominates their design, they typically pair each truck with a single drone [28,40], operate drones independently of trucks [41], or forgo route-level synchronization altogether [42].

Overall, the literature has progressively evolved from conventional truck-based relief distribution toward increasingly integrated truck–drone collaborative delivery systems. However, several limitations remain. First, the simultaneous operation of multiple drones carried by each truck and their parallel delivery missions has received limited attention in emergency supplies dispatch. Second, the joint consideration of flexible drone launch and recovery, truck–drone synchronization, and service waiting remains insufficient. Third, although time-window constraints can ensure that deliveries are completed within specified periods, they do not explicitly distinguish the urgency of different affected sites. In post-disaster relief operations, such differences are important because scarce fast-response resources, such as drones, need to be prioritized for more urgent tasks.

To address these gaps, this study formulates a vehicle routing problem with multiple trucks and multiple drones (VRPMTMD) for emergency supplies dispatch. The model considers multiple drones per truck, flexible launch and recovery, urgency-differentiated delivery deadlines, and a fixed-weighted objective combining transportation cost and delivery time. A hybrid Q-learning-based improved particle swarm optimization and adaptive large neighborhood search algorithm (QL-IPSO-ALNS) is further developed to efficiently solve the proposed problem (Table 1).

3. Problem description and mathematical formulation

3.1. Problem description

The vehicle routing problem with multiple trucks and multiple drones can be described as follows. Consider a set $N = \{1, 2, \cdots, n\}$ of affected sites with delivery demands, where each node $j \in N$ has a demand weight q_j.The distribution center is equipped with s trucks, each of which operates collaboratively with h drones. Each drone must be launched from and recovered by its assigned truck. Every delivery task is executed jointly by the fleet consisting of trucks and drones, and each

Table 1
Literature review on truck-drone coordination in emergency relief distribution.

Ref	MT	MD	CT;CC	FLR	UD	ST
Zhang et al. [35]				√		
Lu et al. [34]	√			√		
Long et al. [36]	√	√		√		√
Van Steenbergen et al. [41]	√					
Yin et al. [40]	√		√	√		
Shi et al. [28]	√		√	√		√
Shi et al. [38]	√			√		
Weng et al. [37]	√			√		√
Guo et al. [39]	√	√	√	√		√
Sun et al. [42]	√					
This paper	√	√	√	√	√	√

Note: In the first row, "Ref", "MT", "MD", "CT;CC", "FLR", "UD", and "ST", denote reference, multi-truck, multi-drone, both time and cost objectives are taken into account, flexible launch and recovery at visited nodes along truck routes, urgency-differentiated delivery deadlines, service time requirement at affected sites.

affected site may be served at most once. After completing all deliveries, all trucks and drones must return to the distribution center. The payload capacity of each truck does not exceed Q_s, and that of each drone does not exceed Q_h. The travel time for a truck between node i and node j is denoted by τ_{ij}^s,while the flight time for a drone between node i and node j is denoted by τ_{ij}^h. Drone endurance is measured in time units and is represented by the parameter δ_h. A drone flight mission can be represented by the triple(i,j,k). Let V_0 and V_+ denote the sets of nodes in the network from which vehicles may depart and to which they may arrive, respectively. The detailed notations are provided in Table 2. (Fig. 1).

Based on the problem context described above, the model is formulated under the following assumptions:

1. The distribution center has s homogeneous trucks, each equipped with h homogeneous drones, with both trucks and drones operating at constant speeds.
2. The launch and recovery times for drones are considered negligible, and the service time at each node is fixed.
3. The supply demand at each affected site is known and must be satisfied in a single delivery.
4. Each affected site is subject to a delivery deadline constraint, permitting early arrival but prohibiting delivery beyond the specified deadline.
5. It is assumed that when a drone is waiting for its truck at a node, the drone is powered off, incurring no waiting cost.
6. The drone charging model adopts battery swapping; sufficient batteries are available on the truck, and the battery swap time is negligible.
7. Drones may launch from and return to a truck only when the truck is stationed at an affected site or at the depot.

Table 2
Notation statement.

	Notation	Description
Sets	$N = \{1,2,\cdots,n\}$	Set of demand nodes
	$V = \{0,1,2,\cdots, n+1\}$	Set of all nodes
	$V_0 = \{0,1,2,\cdots, n\}$	Set of all nodes and start depot 0
	$V_+ = \{1,2,\cdots, n+1\}$	Set of all nodes and the end depot (n+1)
	$S = \{1,2,\cdots,s\}$	Set of trucks
	$H = \{1,2,\cdots,h\}$	Set of drones
Parameters	α	Weight of the distance-related cost in the objective
	β	Weight of the time-related cost in the objective
	c_1	Truck's unit delivery cost
	c_2	Drone's unit delivery cost
	c_3	Unit time cost
	q_j	Demand at node j
	Q_s	Truck's maximum payload capacity
	Q_h	Drone's maximum payload capacity
	δ_h	drones' endurance limitation
	T_s	Completion time of truck s and its drones
	p_i	Urgency priority of site i
	θ	Deadline shrinking coefficient
	b_i	Service time at node i, $b_0 = 0, b_{n+1} = 0$
	d_{ij}^s	Travel distance by truck from node$i$to node$j$
	d_{ij}^h	Flight distance of drone from node$i$to node$j$
	τ_{ij}^s	Truck travel time from node i to node$j$
	τ_{ij}^h	Drone flight time from node i to node$j$
	e_i	Earliest acceptable arrival time of site i
	l_i	Delivery deadline of site i
	M	A sufficiently large positive number
Variables	$x_{ij}^s = \{0,1\}$	Binary variable equal to 1 if the truck travels directly from node$i$to node$j$; 0 otherwise.
	$y_{ijk}^{sh} = \{0,1\}$	Binary variable equal to 1 if the drone is launched from node$i$,serves node$j$,and is recovered at node$k$; 0 otherwise.
	t_i^s	Arrival time of the truck at node$i$
	t_i^{sh}	Arrival time of the drone at node$i$
	r_i^s	Departure time of the truck from node$i$,
	r_i^{sh}	Departure time of the drone from node$i$,
	w_i^s	Waiting time of trucksat node$i$for the drone
	w_i^{sh}	Waiting time of drone$h$at launch node$i$before takeoff
	u_i^s	Visit order of node $i \in N$
	z_i^s	Maximum dwell time of trucks at node$i$
	$\lambda_{ki'}^s = \{0,1\}$	Binary variable equal to 1 if node k is visited before node$i'$by truck s; 0 otherwise;by convention, $\lambda_{0i'}^s = 1$ for all$i' \in N$ and $\lambda_{i0}^s = 0$ for all $i \in N$.

3.2. Mathematical formulation

A mixed-integer linear programming model is developed for the truck–drone collaborative delivery system. To balance transportation efficiency and economic performance in emergency logistics, the objective is to minimize total system cost. As defined in Eq. (1), the objective includes two components: distance-related cost and time-related cost. The former accounts for the cumulative travel distance of all trucks and drones, while the latter reflects the aggregate route completion time across the truck fleet. This two-term structure follows Lu et al. [43], with a key adaptation: whereas Lu et al. model the time component as the makespan in a single-truck setting, the present formulation minimizes the sum of completion times across all trucks. In doing so, it promotes the maximum utilization of the limited fleet under tight vehicle constraints in the critical rescue phase.

$$\min \alpha\left(\sum_{s\in S}\sum_{i\in V}\sum_{j\in V} c_1 d_{ij}^s x_{ij}^s + \sum_{s\in S}\sum_{h\in H}\sum_{i\in V_0}\sum_{j\in N}\sum_{k\in V_+} c_2\left(d_{ij}^h + d_{jk}^h\right) y_{ijk}^{sh}\right) + \beta\left(c_3 \sum_{s\in S} T_s\right) \tag{1}$$

1). General constraints

$$\alpha + \beta = 1 \tag{2}$$

$$\sum_{j\in V_+} x_{0j}^s = 1 \quad \forall s \in S \tag{3}$$

$$\sum_{i\in V_0} x_{in+1}^s = 1 \forall s \in S \tag{4}$$

$$\sum_{\substack{i\in V_0 \\ i\neq j}} x_{ij}^s = \sum_{\substack{k\in V_+ \\ j\neq k}} x_{jk}^s \quad \forall_j \in N, \forall_s \in S \tag{5}$$

$$y_{ijk}^{sh} \le \sum_{l\in V} x_{li}^s \quad \forall i \in V_0, j \in N, k \in V_+, s \in S, h \in H \tag{6}$$

$$y_{ijk}^{sh} \le \sum_{l\in V} x_{lk}^s \quad \forall i \in V_0, j \in N, k \in V_+, s \in S, h \in H \tag{7}$$

$$\sum_{i\in V_0}\sum_{j\in N} y_{ijk}^{sh} \le 1 k \in V_+, \forall_s \in S, h \in H \tag{8}$$

$$\sum_{j\in N}\sum_{k\in V_+} y_{ijk}^{sh} \le 1 i \in V_0, \forall_s \in S, \forall h \in H \tag{9}$$

$$\sum_{s\in S}\sum_{i\in V} x_{ij}^s + \sum_{s\in S}\sum_{h\in H}\sum_{i\in V_0}\sum_{k\in V_+} y_{ijk}^{sh} = 1 \quad \forall j \in N \tag{10}$$

$$y_{ijk}^{sh} = 0 \quad \forall i \in V_0, j \in N, k \in V_+, s \in S, h \in H, i = k \tag{11}$$

$$y_{0jk}^{sh} \le \sum_{\substack{i\in V_0 \\ j\neq i}} x_{ik}^s \quad \forall j \in N, k \in V_+, s \in S, h \in H \tag{12}$$

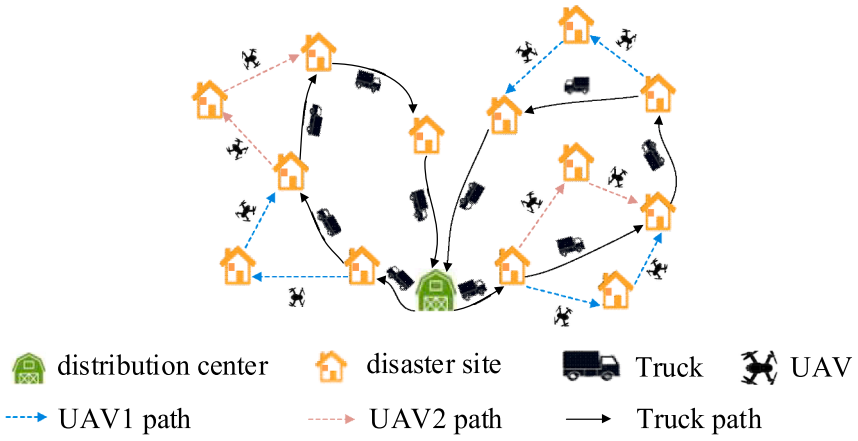


Fig. 1. Schematic illustration of truck-drone collaborative delivery.

$$u_k^s \geq u_i^s + 1 - M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, k \in N, i \neq k, j \in N, s \in S, h \in H \tag{13}$$

Constraint (2) requires that the weights assigned to the distance cost and time cost sum to one. Constraint (3) ensures that each truck departs from the depot along exactly one arc, while Constraint (4) requires each truck to return to the depot along exactly one arc, thereby ensuring route completeness. Constraint (5) imposes flow conservation at each node, requiring the inflow and outflow of each truck to be equal. Constraints (6) and (7) describe the feasibility conditions for collaborative delivery: if a drone is launched from node i and recovered at node k, then both nodes must be visited by trucks. Constraints (8) and (9) ensure that each drone h can be launched at most once from a launch node i and recovered at most once at a recovery node k. Constraint (10) requires each node to be served exactly once. Constraint (11) prevents the launch node and recovery node of a drone from being identical, thereby ensuring that the drone performs an actual delivery mission. Constraint (12) specifies that if a drone is launched directly from the depot to serve a node, the corresponding truck must visit the designated recovery node. Constraint (13) requires the recovery node to be visited after the corresponding launch node.

2). Truck-Drone temporal coordination constraints

$$t_j^s \geq r_i^s + \tau_{ij}^s - M\left(1 - x_{ij}^s\right) \quad \forall i,j \in V, i \neq j, \forall s \in S \tag{14}$$

$$r_i^s \geq t_i^s + b_j \sum_{j \in V} x_{ij}^s - M\left(1 - \sum_{j \in V} x_{ij}^s\right) \forall i \in N, \forall s \in S \tag{15}$$

$$t_i^{sh} + w_i^{sh} \geq r_i^s - M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, \forall h \in H \tag{16}$$

$$t_i^{sh} + w_i^{sh} \leq r_i^s + M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, \forall h \in H \tag{17}$$

$$t_i^{sh} \leq r_i^s - M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, \forall h \in H \tag{18}$$

$$t_j^{sh} \geq r_i^s + \tau_{ij}^h - M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, \forall h \in H \tag{19}$$

$$r_j^{sh} \geq t_j^{sh} + b_j - M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, \forall h \in H \tag{20}$$

$$t_k^{sh} \geq r_j^{sh} + \tau_{jk}^h - M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, \forall h \in H \tag{21}$$

$$r_k^s \geq t_k^{sh} - M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, \forall h \in H \tag{22}$$

$$w_k^s \geq t_k^{sh} - t_k^s - M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, \forall h \in H \tag{23}$$

$$\tau_{ij}^h + b_j + \tau_{jk}^h \leq \delta_h + M\left(1 - y_{ijk}^{sh}\right) \quad \forall i \in V_0, j \in N, k \in V_+, s \in S, h \in H \tag{24}$$

Constraint (14) defines the arrival time of a truck, and Constraint (15) defines its departure time. Constraints (16)–(18) synchronize the launch time of a drone at node i with the departure time of the truck carrying it from that node. Constraint (19) links the drone flight time to the travel time from the launch node to the served node. Constraint (20) requires that the drone departure time from the served node be no earlier than its arrival time plus the corresponding service time. Constraint (21) links the drone return flight time from the served node to the recovery node. Constraint (22) requires the truck to wait at the recovery node until the drone arrives. Constraint (23) defines the truck waiting time at the recovery node. Constraint (24) imposes the drone endurance limit.

3). Capacity and delivery deadline constraints

$$e_i \sum_{j \in V} x_{ji}^s \leq t_i^s \leq l_i \sum_{j \in V} x_{ji}^s + M\left(1 - \sum_{j \in V} x_{ji}^s\right) \quad \forall i \in N, \forall s \in S \tag{25}$$

$$e_j \sum_{i \in V_0} \sum_{k \in V_+} y_{ijk}^{sh} \leq t_j^{sh} \leq l_j \sum_{i \in V_0} \sum_{k \in V_+} y_{ijk}^{sh} + M\left(1 - \sum_{i \in V_0} \sum_{k \in V_+} y_{ijk}^{sh}\right) \quad \forall j \in N, \forall s \in S, \forall h \in H \tag{26}$$

$$\sum_{j \in N} q_j \left(\sum_{i \in V_0} x_{ij}^s + \sum_{h \in H} \sum_{i \in V_0} \sum_{k \in V_+} y_{ijk}^{sh} \right) \leq Q_s \quad \forall s \in S \tag{27}$$

$$q_j y_{ijk}^{sh} \leq Q_h \quad \forall i \in V_0, j \in N, k \in V_+, s \in S, h \in H \tag{28}$$

Constraints (25) and (26) impose delivery deadline constraints on trucks and drones, respectively, ensuring that arrivals at served sites occur within the prescribed deadlines. Constraint (27) imposes the truck capacity constraint, ensuring that the total load assigned to a truck does not exceed its payload capacity. Constraint (28) imposes the drone capacity constraint, limiting the amount of supplies carried by a drone on a single sortie to its payload capacity.

4). Urgency-aware time window constraints

$$\varsigma_i = \frac{p_i - 1}{p_{\max} - 1}, \forall i \in N \tag{29}$$

$$l'_i = \max\{l_i \cdot (1 - \theta \cdot \varsigma_i), e_i + b_i + \varepsilon\} \tag{30}$$

$$e_i \sum_{j \in V} x^s_{ji} \le t^s_i \le l'_i \sum_{j \in V} x^s_{ji} + M\left(1 - \sum_{j \in V} x^s_{ji}\right) \forall i \in N, \forall s \in S \tag{31}$$

$$e_j \sum_{i \in V_0} \sum_{k \in V_+} y^{sh}_{ijk} \le t^{sh}_j \le l'_j \sum_{i \in V_0} \sum_{k \in V_+} y^{sh}_{ijk} + M\left(1 - \sum_{i \in V_0} \sum_{k \in V_+} y^{sh}_{ijk}\right) \quad \forall j \in N, \forall s \in S, \forall h \in H \tag{32}$$

Constraint (29) normalizes the field-assessed rescue priority $p_i \in \{1, \cdots, p_{\max}\}$ into a dimensionless urgency index $\varsigma_i \in [0, 1]$. Constraint (30) derives the effective delivery deadline l_i' by contracting the original deadline l_i in proportion to ς_i, with the lower bound $e_i + b_i + \varepsilon$ ensuring that a feasible service window is preserved. When $\theta = 0$, all effective deadlines reduce to their original values. Constraints (31) and (32) enforce the time window $[e_i, l_i']$ on trucks and drones, respectively, ensuring that every affected site is served within its urgency-adjusted deadline.

5). Drone sequencing constraints

$$\lambda^s_{ki'} = 0 \quad \forall k \subseteq V_0, \forall s \in S \tag{33}$$

$$u^s_k + 1 \le u^s_{i'} + M(1 - \lambda^s_{ki'}) \forall k, i' \in V, k \ne i', \forall s \in S \tag{34}$$

$$u^s_k \ge u^s_{i'} - M \cdot \lambda^s_{ki'} \forall k, i' \in V, k \ne i', \forall s \in S \tag{35}$$

$$t^{sh}_k \le r^s_{i'} + M\left(2 - y^{sh}_{ijk} - y^{sh}_{i'j'k'}\right) + M(1 - \lambda^s_{ii'}) \tag{36}$$

$$\forall i, i' \in V_0, \forall j, j' \in N, \forall k, k' \in V_+, \forall s \in S, \forall h \in H, (i, j, k) \ne (i', j', k')$$

Constraints (33) to (35) define the visit-order indicator λ, and constraint (36) orders multiple missions performed by the same drone according to the visit order of their launch nodes, ensuring that the recovery of an earlier-launched mission is completed no later than the launch of the subsequent one.

6). Total operational time constraints

$$z^s_i \ge 0 \quad \forall i \in V, \forall s \in S \tag{37}$$

$$z^s_i \le M \sum_{j \in V} x^s_{ij} \quad \forall i \in V, \forall s \in S \tag{38}$$

$$z^s_i \ge b_i \sum_{j \in V} x^s_{ij} \quad \forall i \in N, \forall s \in S \tag{39}$$

$$z^s_i \ge w^s_i \quad \forall i \in V, \forall s \in S \tag{40}$$

$$T_s \ge t^{sh}_k \forall i \in V_0, \forall j \in N, \forall k \in V_+, \forall s \in S, h \in H \tag{41}$$

$$T_s \ge \sum_{i \in V} \sum_{j \in V} \tau^s_{ij} x^s_{ij} + \sum_{i \in N} z^s_i \forall s \in S \tag{42}$$

Constraints (37) to (39) impose bounds on the truck dwell-time variable. Constraint (40) ensures that the dwell time at a node is sufficient to cover the truck's waiting time for drone recovery. Constraints (41) and (42) define the total time required for all trucks to complete their delivery tasks and return to the distribution center.

7). Subtour elimination constraints

$$u^s_0 = 0 \quad \forall s \in S \tag{43}$$

$$1 \le u^s_j \le n \quad \forall j \in N, \forall s \in S \tag{44}$$

$$u^s_i - u^s_j + n x^s_{ij} \le n - 1 \forall s \in S, i \ne j, i, j \in N \tag{45}$$

Constraints (43) to (45) eliminate subtours and prevent trucks from revisiting previously visited nodes.

8). Variable domain and non-negativity constraints

$$x^s_{ij} \in \{0, 1\} \forall i, j \in V, i \ne j, \forall s \in S \tag{46}$$

$$y^{sh}_{ijk} \in \{0, 1\} \forall i \in V_0, j \in N, k \in V_+, \forall s \in S, \forall h \in H \tag{47}$$

$$\lambda^s_{k,i'} \in \{0, 1\} \forall k, i' \in V_0, k \ne i', \forall s \in S \tag{48}$$

$$t^s_i \ge 0, r^s_i \ge 0, w^s_i \ge 0 \quad \forall i \in V, \forall s \in S \tag{49}$$

$$t^{sh}_i \ge 0, r^{sh}_i \ge 0, w^{sh}_i \ge 0 \quad \forall i \in V, \forall s \in S, \forall h \in H \tag{50}$$

$$0 \le u^s_i \le n \forall s \in S, i \in N \tag{51}$$

$$z^s_i \ge 0 \quad \forall i \in V, \forall s \in S \tag{52}$$

$$T_s \ge 0, \forall s \in S \tag{53}$$

Constraints (46) to (48) define the binary decision variables. Constraints (49) and (50) specify that the time-related auxiliary variables for trucks and drones, including arrival times, departure times, and waiting times, are non-negative continuous variables. Constraint (51) defines the visit-order variable associated with truck visits to each node. Constraints (52) and (53) specify that the node dwell time variables and total operational time variables are non-negative continuous variables.

4. Solution method

As an extension of the vehicle routing problem (VRP), the VRPMTMD integrates multiple trucks and drones with parallel operations, which increases problem complexity and constraints. Therefore, an efficient heuristic algorithm is required to obtain high-quality solutions. Particle swarm optimization (PSO) is a population-based search method inspired by collective behavior in biological swarms [44]. Due to its simple mechanism, limited number of control parameters, and ease of implementation, PSO has been widely applied in optimization. Nevertheless, when dealing with complex discrete combinatorial problems such as the VRPMTMD, standard PSO tends to converge prematurely and exhibits limited local exploitation capability, which may reduce solution quality and lead to entrapment in local optima [45,46].

The main procedure of the proposed algorithm is outlined in Algorithm 1.

Algorithm 1. Framework of the proposed QL-IPSO-ALNS algorithm

Input: instance I, population size P, maximum number of iterations T_{max}, ALNS interval τ
Output: best solution S^*
1 Randomly initialize the particle population $\{\pi_1, \pi_2, \ldots, \pi_p\}$, Q-table, and algorithm parameters
2 Decode particles and initialize personal bests S_i best and global best S^*
3 $t \leftarrow 0$
4 while t < T_{max} do
5 for $i \leftarrow 1$ to p do
6 Update particle π_i according to the improved PSO strategy
7 $S_i \leftarrow$ Repair (Decode(π_i))
8 Evaluate f(S_i) and update S_i best and S^*
9 end for
10 Update particle categories (elite, explorer, chaser, mutant)
11 if stagnation_count $\ge \theta$ then
12 Reset mutant particles with random encoding
13 end if
14 if (t mod $\tau = 0$ and t > 0) then
15 Select the top 10% particles as elite particles
16 for each elite particle π_i do
17 Select destroy-repair operators using Q-learning
18 Generate and accept a candidate solution using ALNS with simulated annealing
19 Update the Q-table, particle π_i, S_i best, and S^* if necessary
20 end for

(continued on next page)

(*continued*)

Algorithm 1. Framework of the proposed QL-IPSO-ALNS algorithm

```
21 stagnation_count←0
22 end if
23 t←t + 1
24 end while
25 Return:S*
```

To overcome these shortcomings, this study proposes a hybrid QL-IPSO-ALNS algorithm for the VRPMTMD, as illustrated in Fig. 2. The framework combines an improved PSO for global exploration with adaptive large neighborhood search (ALNS) for local intensification. Specifically, problem-oriented encoding and decoding schemes are first designed, together with a tailored initialization method and a fitness-based particle stratification strategy. The ALNS component is then strengthened through dedicated destroy and repair operators, a launch-and-recovery local improvement operator, and a Q-learning-guided adaptive operator selection mechanism. Through this integration, QL-IPSO-ALNS achieves a better balance between exploration and exploitation, thereby improving overall search performance and adaptive search capability. The main procedure of the proposed algorithm is summarized in Algorithm 1.

4.1. Encoding and decoding scheme

A problem-oriented three-layer encoding scheme is designed to represent candidate solutions for the VRPMTMD. For an instance with n affected sites and s trucks, each particle encodes a solution as a vector, $X = \left(x^{(1)}, x^{(2)}, x^{(3)}\right)$ where the three layers capture service priority, service mode, and vehicle assignment, respectively.

The first layer $x_i^{(1)} \in [0,1]$ adopts a continuous random-key representation, where $x_i^{(1)}$ encodes the relative priority of affected site i in the global service sequence. Sorting $x^{(1)}$ in ascending order yields a valid visiting permutation σ over all affected sites, enabling standard PSO velocity–position updates to operate in a continuous space while implicitly defining a discrete routing order. The second layer $x_i^{(2)} \in [-2,2]$ is a continuous variable encoding the service mode of affected site i. A threshold-based decoding rule is applied: if $\lfloor x_i^{(2)} \rfloor = 1$, i.e.,$x_i^{(2)} \in [1,2)$, site i is assigned to drone service; otherwise, it is served directly by the assigned truck. The third layer $x_i^{(3)} \in [1,\mathrm{s}]$ is a continuous variable representing the truck assignment of affected site i. The assigned truck index is obtained by, ensuring the decoded value falls within the valid range $\{1,2\ldots,s\}$.The encoding structure is illustrated in Fig. 3.

The decoding procedure translates the three-layer encoding into a feasible truck–drone collaborative delivery plan through a four-stage

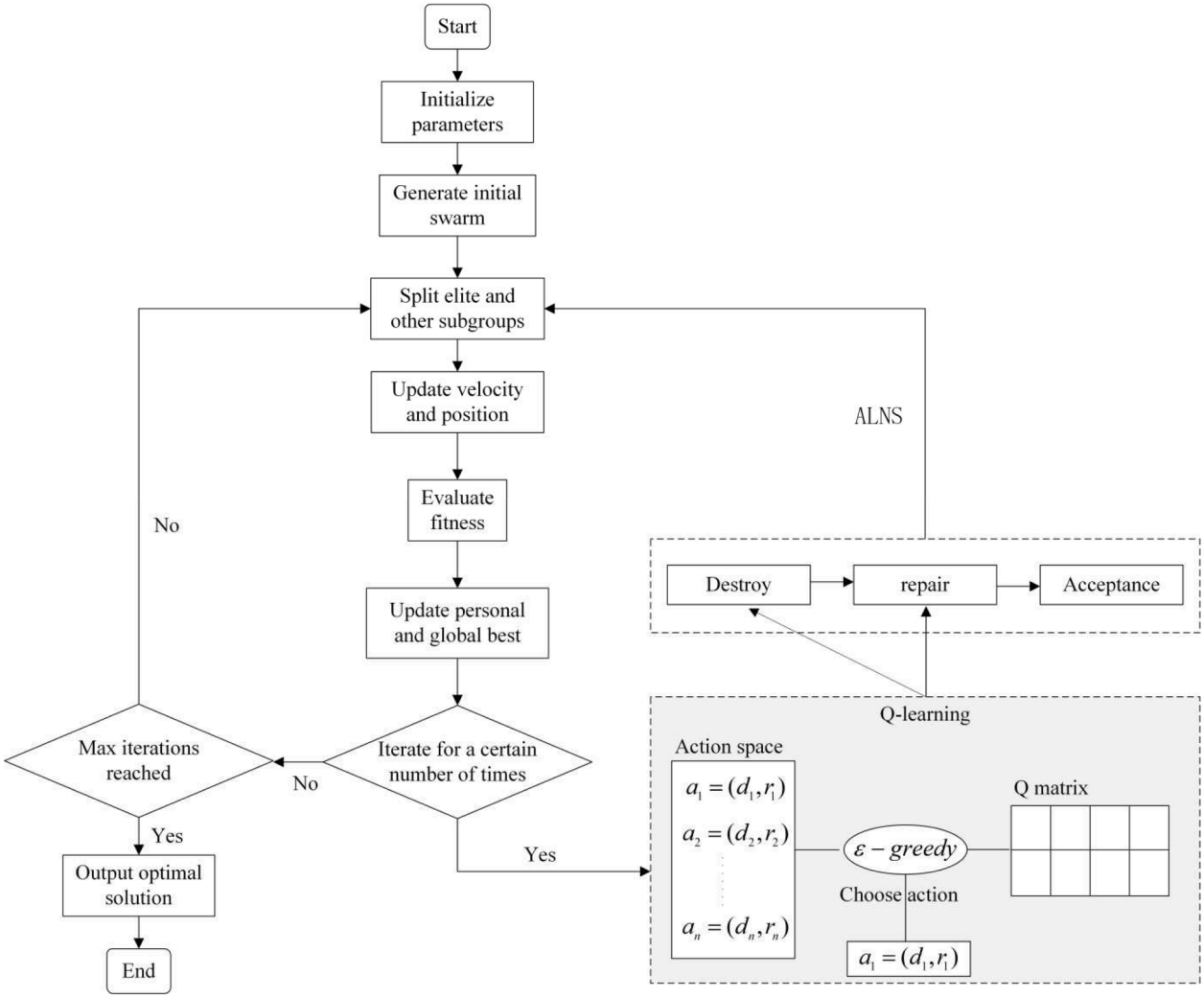


Fig. 2. QL-IPSO-ALNS algorithm framework.

	N_1	N_2	N_3	N_4	N_5	N_6	N_7	N_8	N_9	N_{10}
$x^{(1)}_i$	0.2	0.8	0.1	0.9	0.3	0.5	0.7	0.6	0.4	0.85
$x^{(2)}_i$	−1.2	1.6	−0.6	−1.7	−0.2	0.2	1.3	0.5	−0.5	−1.0
$x^{(3)}_i$	1.2	1.4	2.5	2.1	1.06	2.4	2.7	1.8	1.9	2.03

Fig. 3. Encoding diagram.

hierarchical process, as illustrated in Fig. 4.

Stage 1: Global service sequence generation. The random-key vector $x^{(1)}$ is sorted in ascending order via σ to obtain the global service sequence over all affected sites. This operation transforms continuous particle positions into a valid node permutation suitable for discrete routing optimization.

Stage 2: Service pattern determination. For each affected site i, the truck index and service mode are decoded from $x^{(3)}$ and $x^{(2)}$, respectively. If drone service is indicated but violates the payload or endurance constraints of all available drones on the assigned truck, the site is reassigned to direct truck service to preserve feasibility.

Stage 3: Route and task construction. Sites assigned to truck service are inserted into the corresponding truck routes according to their relative order in σ, constructing round trips originating and terminating at the depot. For sites assigned to drone service, feasible launch–service–recovery triplets (i,j,k) are identified within the truck route, where node i is the launch point, j the served affected site, and k the recovery point. Each triplet must simultaneously satisfy the drone endurance constraint and the truck–drone temporal synchronization constraints. If no feasible triplet exists, the site is reassigned to truck service.

Stage 4: Feasibility repair. After all affected sites have been embedded, a repair procedure eliminates any remaining capacity violations. The final output is a structurally complete and feasible collaborative delivery plan.

4.2. Heuristic initialization strategy

The quality of the initial swarm plays a crucial role in both convergence efficiency and solution quality. Although random initialization ensures diversity, it frequently generates low-quality starting solutions and slows convergence. To mitigate this issue, a heuristic initialization strategy that integrates geographic clustering with nearest-neighbor principles is introduced to improve the overall quality of the initial population.

Step 1. Nodes are first assigned to trucks through a K-means clustering procedure, where affected site coordinates are used as input features and the number of clusters is set equal to the number of trucks. At this stage, capacity constraints are temporarily ignored in order to preserve geographic compactness within each cluster and reduce intra-cluster travel distance. To alleviate the resulting load imbalance, a subsequent balancing mechanism based on edge-node transfer is applied to improve the distribution of demand among clusters.

Step 2. A hierarchical nearest-neighbor strategy is used to construct the affected site service sequence. The visiting order of clusters is first determined, and then a nearest-neighbor rule is applied within each cluster to generate the local affected site sequence. The global service sequence is obtained by concatenating the local sequences according to the cluster order. To avoid overly regular initial solutions and reduce the risk of premature convergence, a mild perturbation is introduced by randomly swapping adjacent affected sites in the resulting sequence.

Step 3. Affected sites are assigned to service modes on the basis of the generated service sequence and truck allocation, while accounting for affected site demand and drone capability. Affected sites that cannot be feasibly served by drones, either because of accessibility restrictions or payload limitations, are assigned directly to truck service. For drone-eligible affected sites, drone service is prioritized, and the task is randomly assigned to an available drone carried by the corresponding truck (Fig. 5).

4.3. Population stratification

Algorithm 2. Particle velocity and position update

Input: Particle π_i, personal best position π_i^{best}, global best position π_g^{best}, inertia weight w, coefficients c_1, c_2, maximum velocity V_{max}
Output: Updated particle π_i
1 Determine the category of πi according to its fitness ranking
2 Set w, c_1, c_2, V_{max} according to the particle category (see Table 3)
3 for d←1 to n do
4 r_1, r_2←*Uniform*$(0,1)$

(*continued on next page*)

Dimension	1	2	3	4	5	6	7	8	9	10
Vehicle index	1	1	2	2	1	2	2	1	1	2
Random-key encoding	0.2	0.8	0.1	0.9	0.3	0.5	0.7	0.6	0.4	0.85
Ascending order	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.85	0.9
Disaster site ID	3	1	5	9	6	8	7	2	10	4
Disaster site ID	1	5	9	8	2	3	6	7	10	4
Vehicle index	1	1	1	1	1	2	2	2	2	2
Truck / UAV	0	2	1	0	0	0	1	2	0	0

Fig.4. Decode diagram.

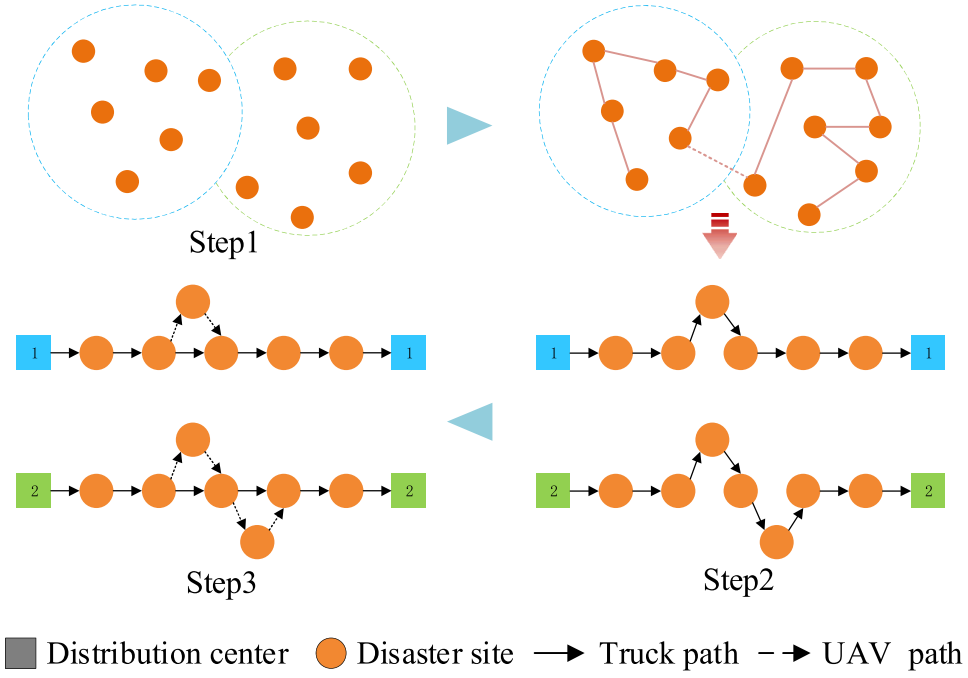


Fig. 5. Heuristic initial solution generation.

(*continued*)

Algorithm 2. Particle velocity and position update

$v_i^{(k)}[d] \leftarrow w \cdot v_i^{(k)}[d] + c_1 r_1 \left(\pi_i^{k.best}[d] - \pi_i^{(k)}[d]\right) + c_2 r_2 \left(\pi_g^{k.best}[d] - \pi_i^{(k)}[d]\right)$, $k = 1,2,3$

$v_i^{(1)}[d] \leftarrow Clamp\left(v_i^{(1)}[d],\ -V_{max},\ V_{max}\right)$

$v_i^{(2)}[d] \leftarrow Clamp\left(v_i^{(2)}[d],\ -10V_{max},\ 10V_{max}\right)$

$v_i^{(3)}[d] \leftarrow Clamp\left(v_i^{(3)}[d],\ -5V_{max},\ 5V_{max}\right)$

end for

$\pi_i^{(1)}[d] \leftarrow Clamp\left(\pi_i^{(1)}[d] + v_i^{(1)}[d],\ 0,\ 1\right)$

$\pi_i^{(2)}[d] \leftarrow \pi_i^{(2)}[d] + v_i^{(2)}[d]$

$\pi_i^{(3)}[d] \leftarrow Clamp\left(\pi_i^{(3)}[d] + v_i^{(3)}[d],\ 0.5,\ s + 0.5\right)$

Return π_i

To improve the search capability of the particle swarm, a fitness-based stratification strategy is introduced. At each iteration, particles are ranked in ascending order of fitness and divided into four groups according to percentile thresholds: elite particles, exploration particles, chaser particles, and mutation particles. Different parameter settings and search roles are assigned to these groups so that the swarm can simultaneously maintain local intensification, regional exploration, convergence pressure, and population diversity. The corresponding velocity and position update procedure is summarized in Algorithm 2.

(1) Elite particles represent the highest-quality solutions in the current swarm. They are assigned a small inertia weight and a large cognitive coefficient, encouraging fine-grained exploitation around their personal best positions. In addition, elite particles are selected for ALNS-based local improvement during the collaboration phase.
(2) Explorer particles maintain a balanced configuration with equal cognitive and social coefficients, supporting both personal experience utilization and global information sharing. Their moderate velocity limit promotes steady diversification across the search space.
(3) Chaser particles are assigned a large social coefficient relative to the cognitive coefficient, biasing their movement toward the global best position. This design accelerates convergence in the middle tier of the population.
(4) Mutant particles are assigned a high inertia weight and low attraction coefficients, granting them large-range, near-random movement. Their primary role is to maintain population diversity. When the consecutive non-improvement counter reaches the stagnation threshold θ, all mutant particles are fully reset with randomly generated encodings, and their personal best records are discarded, forcing exploration of previously unvisited regions of the search space.

Based on the above categorization, each particle type is assigned a distinct set of control parameters, including inertia weight w, cognitive coefficient c_1, social coefficient c_2, and velocity limit v_{max}. These parameter configurations are summarized in Table 3, which ensures that the stratified swarm can effectively balance exploration and exploitation throughout the optimization process.

4.4. Adaptive large neighborhood search

Adaptive Large Neighborhood Search (ALNS) is a widely adopted metaheuristic for combinatorial optimization, with strong performance in routing and scheduling problems [47]. It iteratively destroys and repairs solutions to explore large neighborhoods, while adaptively updating operator selection probabilities based on past performance.

In the proposed framework, ALNS is embedded into the local search phase of PSO to refine particle solutions. For the solution S associated with each particle, ALNS proceeds iteratively as follows. First, based on the current search state, a destroy-repair operator pair (d_i, r_j) is jointly

Table 3
Parameter settings for different particle categories.

Category	Percentile	w	c_1	c_2	v_{max}
Elite	[0%, 10%)	0.3	2.5	1.5	0.10
Explorer	[10%, 40%)	0.5	2.0	2.0	0.20
Chaser	[40%, 70%)	0.5	1.0	2.5	0.25
Mutant	[70%, 100%]	0.9	0.5	0.5	0.40

selected from the action space according to the Q-learning-based adaptive selection strategy described in Section 4.5. Second, the destroy operator d_i is applied to remove a subset of nodes from the current solution, and the repair operator r_j is then applied to reinsert the removed affected sites, thereby generating a new solution S'. Third, the Q-value of the selected operator pair is updated using the reward derived from the improvement achieved by S'. Finally, a simulated annealing acceptance criterion is employed to determine whether S' replaces the current solution S. This process is repeated until the stopping condition is met. The overall procedure is illustrated in Fig. 6.

To adapt ALNS to the studied multi-truck multi-drone collaborative delivery problem, six destroy operators, four repair operators, and one local improvement operator are designed according to the problem structure, aiming to enhance truck–drone coordination and improve local search performance.

4.4.1. Destruction operators

Destroy operators eliminate a subset of affected sites from the current solution to enable solution reconstruction. Let the current solution be denoted by *S,* and let $C_s \subseteq N$ denote the set of affected sites currently served in *S*. Each destroy operator removes q affected sites, producing a removal set $\mathrm{R} \subseteq C_s|\mathrm{R}| = \mathrm{q}$.

(1) Random Removal. This operator randomly selects q affected sites from C_s. For each selected affected site, the corresponding service element is removed from the current solution. If an affected site is served by a truck or a drone, it is removed from the corresponding truck route. If the site also serves as a launch or recovery point for other drone missions, the associated drone service triplets are simultaneously removed.

(2) Worst Cost Removal. For any affected site $i \in C_s$, a composite cost contribution value is defined as

$$D_i = aD_i^D + bD_i^T + D_i^C - D_i^Q \tag{54}$$

Where D_i^D represents the contribution of affected site i to the route distance cost, D_i^T represents its contribution to the time cost, D_i^C denotes the impact on the truck-drone collaborative structure, and D_i^Q represents the capacity value released upon removing the affected site. The parameters a and b are the weighting coefficients for distance cost and time cost in the objective function, respectively. The cost contribution values of all affected sites in the current solution are computed and sorted in descending order of D_i, and the top q affected sites are selected for removal.

(3) Correlation Removal. This operator first selects a seed affected site i from C_s.The correlation between any other affected site $j \in C_s$ and the seed affected site is then evaluated. The remaining $q-1$ affected sites are selected sequentially using a roulette-wheel mechanism based on the correlation measure. This operator is intended to disrupt strongly related local structures so that the repair phase can reorganize groups of similar affected sites more effectively.

(4) Cluster Removal. This operator removes spatially related affected sites from the current solution. A seed affected site i is first selected at random from C_s, and the distances from i to all other affected sites in C_s are computed. The affected sites are then sorted in ascending order of distance, and a neighborhood set of size $\lfloor |C_s| \rfloor = 1$consisting of the closest affected sites is constructed. affected sites are randomly selected from this neighborhood set and added to the removal set R until$|R| \geq q$ or no candidates remain.

(5) Exchange Removal. This operator removes affected sites that may be poorly assigned to their current truck routes. For each affected site, an exchange-reduction score is computed based on its isolation within the current route and its proximity to affected sites on other routes. affected sites with the highest scores, together with their nearest cross-route neighbors, are selected and added to the removal set R until$|R| = q$.

(6) Drone-Optimized Removal. This operator focuses on drone-related tasks. It first collects all drone missions in the current solution and then randomly selects q of them. For each selected mission, only the served affected site is removed, whereas the launch and recovery nodes are retained on the truck route. This operator specifically targets the reallocation of drone-served affected sites without unnecessarily disrupting the truck backbone.

4.4.2. Repair operators

Repair operators reconstruct the partial solution by reinserting the removed nodes in a feasible and cost-effective manner.

(1) Greedy Repair with Truck-Drone Conversion. This operator reconstructs the solution by first inserting the removed affected sites into truck routes and then attempting truck-drone service conversion. The affected sites in R are processed in random order. For each affected site, all feasible insertion positions in all truck

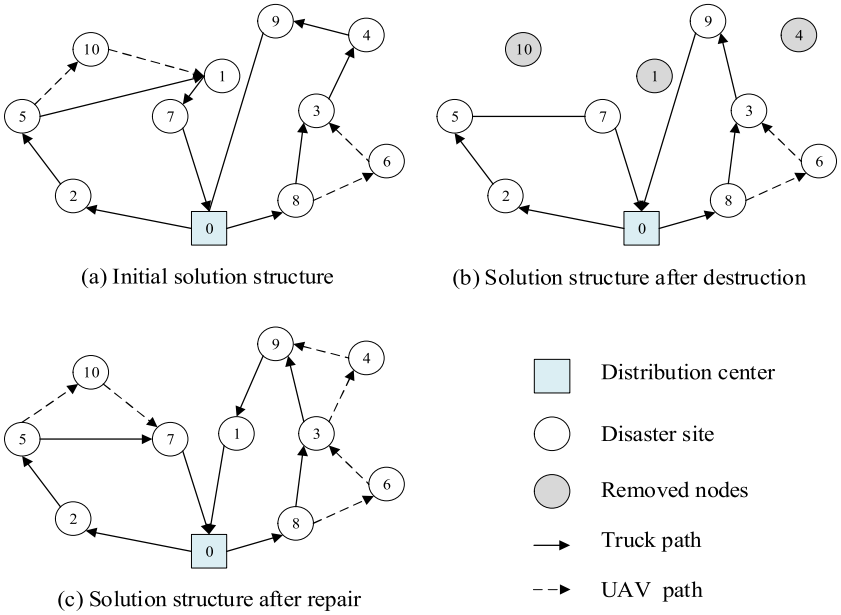


Fig. 6. Illustration of ALNS removal and repair operators.

routes are evaluated, and the position with the smallest increase in the complete objective cost is selected. If no feasible insertion position exists, the affected site is appended to the least-loaded truck route. After all removed affected sites have been inserted, the operator scans truck-served affected sites and attempts to convert selected affected sites to drone service by identifying feasible launch-recovery drone mission triplets. A conversion is accepted only if it reduces the complete objective cost.

(2) Randomized Greedy Repair. This operator introduces controlled randomness into the greedy insertion process to enhance solution diversity. affected sites in the removal set R are first ranked in ascending order of their best insertion costs. At each iteration, one affected site is randomly selected from the first $\lceil L \cdot |R| \rceil$ candidates in the ordered list, where $L \in [0,1]$ is a greediness parameter. The selected affected site is then inserted at its best feasible position. This strategy preserves the efficiency of greedy insertion while reducing the risk of repeatedly constructing similar solutions.

(3) Regret repair. This operator prioritizes affected sites with large regret values, where regret is defined as the additional cost incurred if an affected site is not inserted at the current iteration. In this study, a k-regret insertion rule with $k = 3$ is adopted. For each removed node, the three minimum insertion costs are evaluated. The node with the highest regret is selected and inserted at its best feasible position. This process repeats until all nodes are reinserted.

(4) Closest Insertion Repair. This operator inserts each removed affected site into the route that is spatially closest to it. The affected sites in R are processed in random order. For each affected site, the nearest already served affected site in the current solution is identified, and the truck route containing that affected site is selected as the target route. The operator then evaluates both the best feasible truck insertion and the best feasible drone insertion on the target route. If both are feasible, the lower-cost option is adopted; if only one is feasible, that option is used. affected sites that cannot be inserted by either mode are subsequently handled by the greedy repair procedure.

4.4.3. Launch-and-recovery local improvement operator

Because launch and recovery decisions play a central role in the VRPMTMD, a dedicated local improvement operator is introduced after the ALNS reconstruction step to refine the launch and recovery nodes of drone missions. This operator does not modify affected site assignment or service mode. Instead, it locally adjusts launch and recovery points to reduce flight distance and improve truck–drone coordination efficiency.

4.5. Adaptive destroy-repair operator selection via Q-learning

In the ALNS component of QL-IPSO-ALNS, solution quality depends critically on the selection of destroy-repair operator pairs. Conventional ALNS approaches rely on score-based roulette wheel mechanisms that exhibit limited adaptability to dynamic search contexts, as operator weights are updated reactively over fixed scoring windows without accounting for the current search phase. To address this limitation, a tabular Q-learning mechanism is embedded within ALNS to enable context-aware adaptive operator selection.

4.5.1. Basic Q-learning method

The state–action value function $Q(s, a)$ represents the expected cumulative return of selecting action a in state s. After each iteration, it is updated using the Bellman update rule:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_t + \gamma \max_{a \in A} Q(s_{t+1}, a) - Q(s_t, a_t) \right] \tag{55}$$

where $\alpha = 0.1$ is the learning rate and $\gamma = 0.8$ is the discount factor. At each iteration, action selection follows an ε-greedy policy: with probability ε an operator pair is sampled uniformly at random to sustain exploration; with probability $1 - \varepsilon$ the pair maximizing $Q(s_t, \cdot)$ in the current state is selected, with ties broken randomly. The exploration rate is decayed after each iteration as:

$$\varepsilon \leftarrow \max\{\varepsilon_{\min}, \varepsilon \delta_\varepsilon\} \tag{56}$$

where $\varepsilon_0 = 0.30, \varepsilon_{\min} = 0.10$, and $\delta_\varepsilon = 0.999$. This schedule sustains broad exploration in the early search stage and progressively shifts toward exploitation of high-performing operator combinations as the search matures.

4.5.2. Agent and state

The state space is defined as a four-level discrete set $S = \{0, 1, 2, 3\}$, jointly encoding the current simulated annealing temperature phase and the outcome of the previous ALNS iteration. Let T_t and T_0 denote the current and initial temperatures, respectively. A binary phase indicator ϕ_t is defined as follows: $\phi_t = 1$ if $T_t / T_0 \geq 0.85$, indicating that the search is in the high-temperature exploration phase, and $\phi_t = 0$ otherwise, indicating the low-temperature exploitation phase. Let $b_t \in \{0, 1\}$ indicate whether the previous ALNS iteration improved the incumbent solution, where $b_t = 1$ denotes improvement and $b_t = 0$ denotes no improvement. The state at iteration t is then computed as:

$$S_t = 2(1 - \phi_t) + b_t \tag{57}$$

Accordingly, $s_t = 0$ corresponds to the high-temperature phase with no improvement; $s_t = 1$ to the high-temperature phase with improvement; $s_t = 2$ to the low-temperature phase with no improvement; and $s_t = 3$ to the low-temperature phase with improvement. This compact four-state representation captures the two most informative signals for operator selection: the exploration-exploitation balance reflected by the annealing temperature, and the recent search feedback. Since no historical feedback is available before the first iteration, the initial state is set to $s_0 = 0$.

4.5.3. Action

An action represents the search strategy selected by the learning agent under a given feedback state. In this study, each action is defined as an ordered destroy–repair operator pair. Let D and R denote the sets of destroy and repair operators, respectively; the action space is therefore given by $A \subseteq D \times R$, where each action $a \in A$ takes the form $a = (d, r)$, $d \in D, r \in R$. At each ALNS iteration, the agent selects one operator pair $a_t = (d_t, r_t)$ according to the current state and the Q-table, and the selected destroy and repair operators are then applied sequentially to generate a candidate solution. Modeling the operator pair as a single action enables the learning process to capture the coupled effect between perturbation and reconstruction, rather than assessing destroy and repair operators independently.

4.5.4. Reward

After each ALNS iteration, a continuous reward signal is computed to evaluate the selected operator pair. Let f^{old} and f^{new} denote the incumbent and accepted objective values at the start and end of the iteration, and let f^*_{prev} and f^*_{curr} denote the global best values before and after the iteration. The local and global improvement ratios are defined as:

$$\Delta_{local} = \max\left(\frac{f^{old} - f^{new}}{f^{old}}, 0\right), \Delta_{global} = \max\left(\frac{f^*_{prev} - f^*_{curr}}{f^*_{prev}}, 0\right) \tag{58}$$

The composite improvement is $imp = \eta \cdot \Delta_{global} + (1 - \eta) \cdot \Delta_{local}$, where $\eta = 0.1$ assigns primary weight to local improvement. The reward is then defined as:

$$r_t = \begin{cases} \min\left(\dfrac{imp}{\max(\rho, 0.001)}, 1.0\right), imp > 0 \\ -0.5, otherwise \end{cases} \quad (59)$$

where ρ is a moving-average baseline updated as $\rho \leftarrow 0.9\rho + 0.1 \cdot imp$, normalizing rewards across different problem scales. An additional penalty of -2.0 is applied whenever the generated solution is infeasible.

5. Computational experiments

To assess the effectiveness and robustness of the proposed method, experiments are performed on multiple benchmark instance sets, with each instance tested 10 independent times to ensure result reliability. The Solomon benchmark instances are used to generate test problems of different sizes. For small- and medium-scale instances, both Gurobi and QL-IPSO-ALNS are applied in order to verify the correctness of the mathematical model and assess the solution quality of the proposed algorithm against exact solutions. For large- scale instances, QL-IPSO-ALNS is compared with two benchmark heuristics, namely IPSO and K-means-ALNS. All algorithms are implemented in Python 3.12 and executed on a computer equipped with an Intel(R) Core (TM) i5-12400 CPU @ 2.50 GHz, 16 GB RAM, and a Windows operating system.

5.1. Parameter and instance configuration

To evaluate the proposed algorithm, six instance classes from the Solomon CVRP benchmark are used: C1, C2, R1, R2, RC1, and RC2. In this benchmark, C denotes clustered affected sites, R denotes randomly distributed affected sites, and RC denotes a mixed spatial distribution. The test set includes small-scale instances with 6, 8, and 10 affected sites, medium-scale instances with 15, 20, and 25 affected sites, and large-scale instances with 30, 50, 75, and 100 affected sites. In instance construction, the coordinates of demand nodes are directly adopted from the original Solomon instances, and the demand quantities at each affected site are taken directly from the original instances. The time-window parameters are derived from the earliest service time e_i and deadline $l_i$specified in the original instances. On this basis, a rescue priority $p_i \in \{1, \ldots, p_{max}\}$ is randomly generated for each affected site, and the urgency-aware deadline adjustment is applied with a shrinking coefficient of $\theta = 0.10$.

Following Lu et al. [43], the model parameters are summarized in Table 4. Each affected site requires a fixed service time of 10 min. The cost structure is defined as 4 RMB/km for trucks and 1 RMB/km for drones, with a time cost of 1 RMB/min. In the objective function, the weighting factors are set to $\alpha = 0.7$ for cost and $\beta = 0.3$ for time. The algorithmic parameters are configured as follows: the population size P is scaled according to the problem size and is set to 30, 40, 55, and 70, respectively, the maximum number of iterations is Tmax = 40, and the elite particles are cooperatively improved by the ALNS procedure every $\tau = 10$ iterations. We further conduct the Wilcoxon rank-sum test between QL-IPSO-ALNS and the benchmark algorithms at a significance level of 0.05.

Table 4
Relevant variables setting.

Parameter	Value
Truck Monetary cost	4 RMB/km
Drone Monetary cost	1 RMB/km
Time-related costs	1 RMB/min
Truck load capacity	200 kg
Drone load capacity	15 kg
Drone endurance	30 min
Drone flight speed	60 km/h
Truck travel speed	30 km/h

5.2. Analysis of small- and medium-scale instances

To evaluate the effectiveness of QL-IPSO-ALNS, its performance is first compared with exact solutions obtained by Gurobi. Because the problem is NP-hard, exact solvers become increasingly time-consuming as the problem size grows. In this subsection, Gurobi is applied to small-scale instances according to the proposed mixed-integer programming model, and the resulting optimal solutions are used as a benchmark for evaluating the heuristic algorithm.

The results for the small-scale instances are reported in Table 5. It presents the optimal solutions and computation times obtained by Gurobi, together with the best solutions, average solutions, average computation times, and relative gaps obtained by QL-IPSO-ALNS over multiple independent runs. In terms of solution quality, QL-IPSO-ALNS reaches the same optimal objective value as Gurobi for most instances. Specifically, for C101_6, C101_8, R101_6, R101_8, R101_10, RC101_6, RC101_8, and RC101_10, the optimal objective values obtained by the two methods are identical, yielding a gap of 0.00%. For C101_10, the gap is only 0.15%.

$$GAP = Opt_{QL-IPSO-ALNS} - Opt_{Gurobi}.$$

$$gap = \frac{GAP}{Opt_{Gurobi}} * 100\%.$$

In terms of computational efficiency, QL-IPSO-ALNS requires far less computation time than Gurobi across all small-scale instances. For example, on instance C101_10, Gurobi does not converge within the 3600 s time limit, whereas QL-IPSO-ALNS obtains a near-optimal solution in approximately 0.56 s. This result suggests that the clustered structure of C-type instances may induce a larger number of drone-feasible assignments, thereby increasing the complexity of the exact model.

To further assess the performance of the algorithm as the problem size increases, medium-scale instances are tested, with the results reported in Table 6. For all such instances, Gurobi reaches the 3600 s time limit, highlighting the rapid growth in computational complexity. By contrast, QL-IPSO-ALNS consistently produces high-quality solutions within short computation times, ranging from 0.31 s to 2.16 s on average. In several instances, the solutions found by QL-IPSO-ALNS match the optimal solutions reported by Gurobi, and for the remaining instances the optimality gap never exceeds 0.55%.

Overall, the results on small- and medium-scale instances indicate that QL-IPSO-ALNS provides high-quality solutions while dramatically reducing computation time. Even as the problem size increases, the proposed algorithm remains capable of producing optimal or near-optimal solutions within a practically negligible runtime.

5.3. Analysis of large-scale instances

The VRPMTMD becomes substantially more difficult as the number of affected sites increases because both the problem dimension and the corresponding solution space grow rapidly. As shown in Section 5.2, once the number of affected sites reaches 25, Gurobi already encounters serious computational difficulty. Therefore, for large-scale instances, QL-IPSO-ALNS is first compared with IPSO and K-means-ALNS on the Solomon-derived instances. Additional comparisons with LNS, CH-SaBO, and MAAGLNS are then conducted under a unified experimental setting to further evaluate the search capability of the proposed algorithm. Each instance is solved independently 10 times. Table 7 presents the optimal value (Opt), average objective value (Ave), and relative deviation (Gap) for each algorithm. To quantify the performance differences relative to QL-IPSO-ALNS, two relative deviation measures, Gap1 and Gap2, are further defined for the best and average objective values, respectively.

$$GAP1 = Opt_{IPSO/K-means-ALNS} - Opt_{QL-IPSO-ALNS}$$

$$Gap1 = \frac{GAP1}{Opt_{IPSO/Kmeans-ALNS}} * 100\% \quad ,$$

Table 5
Experimental results on small-scale instances.

Instance	Gurobi		QL-IPSO-ALNS				
	Opt	Time(s)	Opt	Ave	Time(s)	std	gap(%)
C101_6	172.68	6.46	172.68	172.68	0.26	0.00	0.00%
C101_8	185.25	512	185.25	185.50	0.48	0.40	0.00%
C101_10	194.39	3600.00	194.69	195.63	0.56	1.04	0.15%
R101_6	424.69	0.23	424.69	424.69	0.21	0.00	0.00%
R101_8	478.70	3.88	478.70	478.70	0.23	0.00	0.00%
R101_10	574.89	6.70	574.89	574.89	0.23	0.00	0.00%
RC101_6	318.12	0.31	318.12	318.12	0.20	0.00	0.00%
RC101_8	344.05	20.72	344.05	344.05	0.36	0.00	0.00%
RC101_10	588.61	2105.83	588.61	588.61	0.38	0.00	0.00%

Opt denotes the best (optimal) solution, while Ave denotes the average solution value.

Table 6
Experimental results on medium-scale instances.

Instance	Gurobi		QL-IPSO-ALNS				
	Opt	Time(s)	Opt	Ave	Time(s)	std	gap(%)
C101_15	485.41	3600.00	487.65	491.46	0.85	2.46	0.46%
C101_20	568.45	3600.00	568.45	571.46	1.16	7.62	0.00%
C101_25	666.77	3600.00	670.43	677.12	1.19	7.41	0.55%
R101_15	881.94	3600.00	881.94	889.86	0.31	13.25	0.00%
R101_20	1050.29	3600.00	1051.57	1065.62	1.60	24.28	0.12%
R101_25	1191.47	3600.00	1192.74	1218.19	2.16	20.39	0.11%
RC101_15	667.73	3600.00	667.73	668.35	0.67	3.12	0.00%
RC101_20	1012.10	3600.00	1012.10	1012.74	0.74	2.92	0.00%
RC101_25	1061.10	3600.00	1061.10	1065.76	1.24	2.62	0.00%

Opt denotes the best (optimal) solution, while Ave denotes the average solution value.

Table 7
Calculation results of large-scale instances.

Instance	QL-IPSO-ALNS			IPSO				K-means-ALNS			
	Opt	Ave	Time(s)	Opt	Ave	Gap1	Gap2	Opt	Ave	Gap1	Gap2
C101_30	720.43	730.00	2.48	733.12	743.26	1.73%	1.78%	724.13	748.57	0.51%	2.48%
C102_30	724.09	735.83	2.96	735.56	742.88	1.56%	0.95%	736.87	746.29	1.73%	1.40%
C101_50	1295.00	1300.95	19.69	1331.39	1343.30	2.73%	3.15%	1293.29	1351.93	-0.13%	3.77%
C102_50	1273.38	1293.02	19.03	1329.44	1342.98	4.22%	3.72%	1302.17	1341.73	2.21%	3.63%
C101_75	2320.82	2339.73	72.91	2399.90	2411.48	3.30%	2.98%	2376.91	2453.66	2.36%	4.64%
C102_75	2330.80	2359.24	77.04	2402.58	2413.48	2.99%	2.25%	2407.96	2421.66	3.20%	2.58%
C101_100	2976.48	3014.66	325.21	3083.07	3096.32	3.46%	2.64%	3030.37	3128.20	1.78%	3.63%
C102_100	2995.28	3012.9	290.96	3084.17	3101.46	2.88%	2.86%	3046.91	3167.86	1.69%	4.89%
R101_30	1367.81	1393.93	1.22	1395.98	1408.6	2.02%	1.04%	1371.01	1431.17	0.23%	2.60%
R102_30	1358.69	1391.84	1.16	1392.00	1412.22	2.39%	1.44%	1359.86	1437.02	0.09%	3.14%
R101_50	1900.01	1933.59	8.57	2010.81	2070.87	5.51%	6.63%	1940.67	2051.87	2.10%	5.76%
R102_50	1888.56	1996.65	8.69	2002.58	2077.84	5.69%	3.91%	1957.16	2054.76	3.51%	2.83%
R101_75	2695.38	2758.33	51.53	2861.91	2922.09	5.82%	5.60%	2782.16	2957.30	3.12%	6.73%
R102_75	2642.99	2768.56	52.99	2878.82	2947.99	8.19%	6.09%	2778.53	3001.38	4.88%	7.76%
R101_100	3221.90	3334.57	262.48	3464.63	3642.30	7.01%	8.45%	3390.74	3745.13	4.98%	10.96%
R102_100	3258.06	3412.87	241.01	3502.79	3635.15	6.99%	6.11%	3455.15	3888.54	5.70%	12.23%
RC101_30	1488.19	1488.19	2.92	1489.39	1501.62	0.08%	0.89%	1488.19	1504.33	0.00%	1.07%
RC102_30	1488.19	1490.48	2.75	1493.19	1499.65	0.33%	0.61%	1488.19	1497.01	0.00%	0.44%
RC101_50	1879.34	1885.11	21.21	1914.73	1929.82	1.85%	2.32%	1888.99	1911.61	0.51%	1.39%
RC102_50	1879.34	1879.74	19.41	1903.17	1915.25	1.25%	1.85%	1882.57	1919.41	0.17%	2.07%
RC101_75	2994.24	3040.43	61.67	3145.79	3164.76	4.82%	3.93%	3196.23	3305.29	6.32%	8.01%
RC102_75	3031.32	3054.99	71.72	3131.51	3156.75	3.20%	3.22%	3146.87	3384.11	3.67%	9.73%
RC101_100	3563.6	3682.66	187.14	3877.82	4032.95	8.10%	8.69%	3841.05	4295.41	7.22%	14.27%
RC102_100	3568.17	3774.05	172.85	3872.38	4026.01	7.86%	6.26%	3889.00	4313.34	8.25%	12.50%

Opt denotes the best (optimal) solution, while Ave denotes the average solution value.

$$GAP2 = Ave_{Ipso/K-means-ALNS} - Ave_{QL-IPSO-ALNS}$$

$$Gap2 = \frac{GAP2}{Ave_{IPSO/Kmeans-ALNS}} * 100\%$$

As shown in Table 7, the performance differences among the algorithms become increasingly pronounced as the instance size grows from 30 to 100 affected sites. QL-IPSO-ALNS generally obtains the best solutions and consistently lower average objective values. For 30-affected-site instances, the improvements in the best objective value range from 0.08% to 2.39% compared with IPSO, while gains of up to 1.73% are achieved over K-means-ALNS. As the problem size increases to 50 affected sites, the improvement over IPSO expands to 1.25%–5.69%, and reaches up to 3.51% over K-means-ALNS. For 75-affected-site instances, the superiority of QL-IPSO-ALNS becomes more pronounced, yielding improvements of 2.99%–8.19% and 2.36%–6.32% over IPSO and K-means-ALNS, respectively. Similar trends are observed for 100-

affected-site instances, where the proposed algorithm achieves improvements ranging from 2.88% to 8.10% over IPSO and from 1.69% to 8.25% over K-means-ALNS. These results indicate that the proposed algorithm scales effectively with increasing problem size and maintains strong optimization capability for large-scale emergency logistics routing problems.

With respect to average solution quality, QL-IPSO-ALNS also shows consistently better performance. Its average deviation ranges from 0.46% to 1.62% for C-type instances, 1.77% to 5.72% for R-type instances, and 0.00% to 5.77% for RC-type instances, all substantially lower than those of the benchmark heuristics. In all cases, these values are substantially lower than those achieved by the benchmark heuristics. By contrast, IPSO and K-means-ALNS exhibit overall average deviation ranges of approximately 3%–5% and 4%–6%, respectively, and the difference becomes more evident as the problem size increases. In particular, for 100-node RC instances, the average deviation of K-means-ALNS remains the highest among the compared methods, indicating limited robustness in complex mixed-distribution scenarios.

The better performance of QL-IPSO-ALNS on R-type and RC-type instances is attributed to their greater spatial heterogeneity. Compared with C-type instances, in which affected sites are relatively concentrated and high-quality routing patterns are easier to identify, R-type and RC-type instances involve more dispersed and irregular spatial distributions. This increases the size of the solution space and complicates both routing decisions and truck-drone coordination. In this context, conventional heuristics are more prone to premature convergence or insufficient neighborhood exploration. QL-IPSO-ALNS mitigates these difficulties by combining the diversification capability of improved particle swarm optimization with the intensification mechanism of adaptive large neighborhood search. Specifically, IPSO enhances exploration of promising regions in the solution space, whereas ALNS improves solution quality through adaptive destroy-and-repair operations. Therefore, the hybrid framework is particularly effective for instances with strong spatial heterogeneity, which can explain its superior performance on R-type and RC-type instances.

To further verify the effectiveness of the proposed QL-IPSO-ALNS algorithm, comparative experiments are conducted with the large neighborhood search (LNS) of Kitjacharoenchai et al. [11], the CH-SaBO algorithm proposed by Zhai et al. [48], and the MAAGLNS algorithm developed by Chen et al. [49]. These methods provide representative benchmarks with different search mechanisms for evaluating the effectiveness of the proposed approach. To ensure a fair comparison, all algorithms are evaluated under the same experimental settings. Specifically, the objective function is unified to consider only the overall delivery time, while delivery time windows, service times, and urgency-aware delivery deadlines are excluded from this comparative experiment. This setting eliminates the influence of model-specific features and allows the performance differences to be attributed primarily to the search strategies of the algorithms. The same benchmark instances, fleet configuration, and basic operational constraints are applied to all algorithms. The comparative results are presented in Tables 8 and 9.

The results in Table 8 show that QL-IPSO-ALNS consistently achieves competitive performance against LNS across the benchmark instances. Overall, QL-IPSO-ALNS obtains average reductions of 3.45% and 4.91% in the best and average objective values, respectively. The improvement is particularly evident for several medium- and large-scale instances. For example, the reduction in the best objective value reaches 8.74% for E-n76-k10 and 7.94% for E-n76-k8, while the corresponding reduction in the average objective value reaches 8.43% for B-n50-k7 and 8.42% for B-n51-k7. These results indicate that the advantage of QL-IPSO-ALNS becomes more apparent as the problem size and routing complexity increase.

Although LNS occasionally obtains a slightly better best-found solution on individual instances, the negative Gap1 values are limited and do not change the overall comparison. More importantly, QL-IPSO-ALNS maintains a stronger performance in terms of average solution quality, indicating better stability across independent runs. This improvement can be attributed to the integration of Q-learning into the ALNS framework. Instead of selecting destroy-and-repair operators according to fixed or manually designed probabilities, Q-learning dynamically adjusts operator selection according to the current search state and the historical performance of different operator combinations. This mechanism allows the algorithm to identify more promising neighborhood

Table 8
Comparative results of QL-IPSO-ALNS and LNS.

Instance	QL-IPSO-ALNS				LNS				Gap1	Gap2
	Opt	Ave	Std	Time(s)	Opt	Ave	Std	Time(s)		
A-n32-k5	1587.39	1595.01	14.76	7.22	1610.82	1644.20	47.43	10.08	1.45%	2.99%
A-n33-k5	1307.47	1317.79	11.11	8.03	1323.11	1338.56	25.55	10.05	1.18%	1.55%
A-n34-k5	1579.45	1602.28	20.09	9.03	1618.17	1700.79	95.50	10.03	2.39%	5.79%
A-n36-k5	1569.40	1640.25	68.52	9.28	1655.12	1669.76	9.17	10.40	5.18%	1.77%
A-n37-k5	1306.15	1355.80	30.17	9.48	1301.12	1375.3	60.01	10.26	−0.39%	1.42%
A-n38-k6	1454.40	1484.89	16.96	8.85	1449.21	1482.87	45.28	10.40	−0.36%	−0.14%
A-n39-k6	1610.90	1655.40	26.78	7.69	1665.12	1700.19	32.48	10.05	3.26%	2.63%
A-n44-k6	1840.92	1907.77	27.18	11.01	1955.27	2049.25	86.35	15.05	5.85%	6.90%
A-n45-k7	2295.49	2326.16	29.15	7.30	2397.79	2499.06	70.44	15.07	4.27%	6.92%
A-n46-k7	1802.42	1855.52	52.50	13.32	1871.46	1978.63	75.72	15.11	3.69%	6.22%
B-n45-k5	1546.14	1594.96	47.63	13.18	1592.34	1707.14	118.04	15.04	2.90%	6.57%
B-n50-k7	1439.37	1467.33	18.33	12.66	1504.91	1602.35	114.25	25.05	4.36%	8.43%
B-n50-k8	2658.06	2688.60	20.53	13.60	2738.86	2826.90	90.74	25.08	2.95%	4.89%
B-n51-k7	2261.78	2340.22	119.59	13.48	2368.65	2555.34	157.69	25.15	4.51%	8.42%
B-n52-k7	1468.16	1500.97	30.18	17.08	1513.98	1625.71	134.51	25.10	3.03%	7.67%
B-n56-k7	1373.03	1397.84	20.49	16.15	1436.94	1519.95	148.35	50.17	4.45%	8.03%
B-n57-k9	3260.47	3328.38	53.00	28.51	3309.40	3397.25	74.43	50.22	1.48%	2.03%
B-n63-k10	3073.13	3134.71	62.68	20.47	3131.09	3208.41	46.82	100.21	1.85%	2.30%
E-n76-k7	1461.33	1540.69	82.13	26.24	1533.67	1663.79	87.36	120.38	4.72%	7.40%
E-n76-k8	1422.32	1467.67	38.05	38.65	1545.04	1565.01	13.55	120.38	7.94%	6.22%
E-n76-k10	1756.30	1853.03	75.76	87.47	1924.60	1995.36	52.63	150.26	8.74%	7.13%
E-n101-k8	1686.23	1754.38	62.87	87.64	1727.95	1804.83	71.36	200.33	2.41%	2.80%
Wilcoxon (+/≈/−)	–	–	–	–	–	18-4-0	–	–	-	-
Average	–	–	–	–	–	–	–	–	3.45%	4.91%

Note: Based on the Wilcoxon rank-sum test, symbols (+/≈/−) denote that QL-IPSO-ALNS is significantly better, statistically tied, or worse with the comparison algorithm, respectively.

Table 9
Comparative results of QL-IPSO-ALNS with CH-SaBO and MAAGLNS.

Instance	CH-SaBO				Gap1	Gap2	MAAGLNS				Gap1	Gap2
	Opt	Ave	Std	Time(s)			Opt	Ave	Std	Time(s)		
A-n32-k5	1664.64	1682.97	22.97	10.09	4.64%	5.23%	1579.13	1639.33	37.25	10.01	−0.52%	2.70%
A-n33-k5	1317.84	1333.55	17.84	10.18	0.79%	1.18%	1332.37	1351.43	22.02	10.01	1.87%	2.49%
A-n34-k5	1648.27	1695.26	70.59	10.08	4.18%	5.48%	1588.74	1604.26	10.93	10.02	0.58%	0.12%
A-n36-k5	1716.26	1752.30	59.92	10.07	8.56%	6.39%	1687.32	1701.08	12.23	10.01	6.99%	3.58%
A-n37-k5	1439.70	1484.74	49.18	10.13	9.28%	8.68%	1328.73	1368.35	30.70	10.03	1.70%	0.92%
A-n38-k6	1514.48	1582.07	121.40	10.22	3.97%	6.14%	1448.51	1491.61	34.93	10.01	−0.41%	0.45%
A-n39-k6	1670.50	1675.33	7.83	10.2	3.57%	1.19%	1669.17	1717.85	39.83	11.01	3.49%	3.64%
A-n44-k6	1878.92	1977.10	74.35	15.12	2.02%	3.51%	1977.84	2019.06	33.89	15.01	6.92%	5.51%
A-n45-k7	2349.53	2400.55	33.90	15.16	2.30%	3.10%	2405.83	2441.79	30.38	15.02	4.59%	4.74%
A-n46-k7	1911.05	2038.70	120.56	15.46	5.68%	8.99%	1892.02	1910.96	21.52	15.01	4.74%	2.90%
B-n45-k5	1569.98	1674.88	69.19	15.31	1.52%	4.77%	1535.09	1564.79	26.95	15.01	−0.72%	−1.93%
B-n50-k7	1486.15	1515.94	52.14	25.27	3.15%	3.21%	1492.14	1518.53	20.85	25.01	3.54%	3.37%
B-n50-k8	2846.88	2980.19	103.74	25.07	6.63%	9.78%	2671.31	2696.66	27.55	25.01	0.50%	0.30%
B-n51-k7	2299.25	2675.12	410.13	25.38	1.63%	12.52%	2282.71	2785.34	480.76	25.01	0.92%	15.98%
B-n52-k7	1458.60	1532.29	66.92	25.13	−0.66%	2.04%	1553.90	1621.81	38.14	25.01	5.52%	7.45%
B-n56-k7	1391.71	1396.74	7.53	50.18	1.34%	−0.08%	1439.07	1460.06	13.73	50.04	4.59%	4.26%
B-n57-k9	3309.39	3380.20	66.97	50.64	1.48%	1.53%	3278.89	3341.12	24.24	50.01	0.56%	0.38%
B-n63-k10	3272.01	3330.32	48.17	50.75	6.08%	5.87%	3357.94	3396.36	29.31	51.01	8.48%	7.70%
E-n76-k7	1557.09	1604.65	49.16	120.69	6.15%	3.99%	1523.96	1586.42	27.28	120.02	4.11%	2.88%
E-n76-k8	1518.56	1556.83	25.46	120.26	6.34%	5.73%	1438.70	1472.94	22.36	120.18	1.14%	0.36%
E-n76-k10	1953.26	2013.67	103.48	120.33	10.08%	7.98%	1826.11	1937.59	64.31	120.02	3.82%	4.36%
E-n101-k8	1732.89	1870.37	51.02	200.18	2.69%	6.20%	1785.86	1835.62	59.98	200.02	5.58%	4.43%
Wilcoxon (+/≈/−)	–	20-2-0	–	–	–	–	–	16-6-0	–	–	-	-
Average	–	–	–	–	4.16%	5.30%	–	–	–	–	3.09%	3.48%

structures and allocate search effort more effectively, thereby improving both solution quality and search robustness.

Table 9 further compares QL-IPSO-ALNS with CH-SaBO and MAAGLNS, providing a broader evaluation against recently developed heuristic methods. The results show that QL-IPSO-ALNS achieves a generally better overall performance, although the magnitude of improvement varies across different instance structures. Compared with CH-SaBO and MAAGLNS, the proposed algorithm obtains lower best and average objective values in most benchmark instances, while maintaining relatively stable performance across repeated runs. These results demonstrate that the proposed hybrid framework remains competitive when compared with algorithms based on different search strategies.

The advantage becomes more pronounced when QL-IPSO-ALNS is compared with CH-SaBO. On average, QL-IPSO-ALNS reduces the best and average objective values by 4.16% and 5.30%, respectively. The improvement is particularly substantial for several challenging instances. For example, Gap1 reaches 10.08% for E-n76-k10, 9.28% for A-n37-k5, and 8.56% for A-n36-k5. In terms of average solution quality, Gap2 reaches 12.52% for B-n51-k7 and 9.60% forA-n36-k5. These results suggest that QL-IPSO-ALNS is more effective in handling complex combinations of truck routing, drone assignment, and launch–recovery synchronization. Such coupled decisions create strong interactions between the truck and drone routes, making it difficult for a single search mechanism to efficiently explore the solution space. The proposed framework addresses this difficulty by combining the global exploration capability of IPSO with the targeted neighborhood refinement of ALNS, while Q-learning adaptively guides the selection of destroy-and-repair operators according to the current search state.

MAAGLNS provides a more competitive benchmark because it also incorporates adaptive neighborhood search and therefore has a stronger local optimization capability than conventional heuristic methods. Nevertheless, QL-IPSO-ALNS still achieves average reductions of 3.09% and 3.48% in the best and average objective values, respectively. Several instances exhibit particularly large improvements. For B-n63-k10, QL-IPSO-ALNS reduces the best and average objective values by 8.48% and 7.70%, respectively, while the improvement in the average objective value reaches 15.98% for B-n51-k7. These results indicate that the proposed algorithm maintains a clear advantage in terms of average solution quality even against a competitive adaptive-neighborhood-based method.

It is also worth noting that the performance differences between QL-IPSO-ALNS and the benchmark algorithms are relatively limited for some instances, as reflected in the relatively large number of statistical ties (=) in the Wilcoxon rank-sum test. This may be partly attributed to differences in drone utilization strategies. The solutions obtained by MAAGLNS tend to deploy drones less frequently and rely more heavily on trucks. For the sake of a fair comparison, the objective function in our experiments is defined to minimize total delivery time. Although drones are faster than trucks, their speed advantage is relatively modest, while trucks need to wait for and recover drones before continuing their routes. When affected sites are close to truck routes or the potential distance savings from drone deployment are limited, these factors may offset the time savings of drones. Therefore, different routing and drone-assignment strategies may still result in similar total delivery times in some instances.

5.4. Ablation study on the Q-learning-based operator selection strategy

To assess the effectiveness of the adaptive operator selection mechanism embedded in QL-IPSO-ALNS, comparative experiments are conducted between QL-IPSO-ALNS and RW-IPSO-ALNS, a variant in which the adaptive learning mechanism is removed and operators are selected using a roulette-wheel mechanism.

The results in Table 10 show that QL-IPSO-ALNS achieves better performance in most benchmark instances, QL-IPSO-ALNS achieves average improvements of approximately 0.52% in the best solution value and 0.88% in the average solution value. Although the magnitude of improvement is relatively moderate, the superiority is consistently observed across different instance classes, demonstrating the effectiveness of the Q-learning-based operator selection strategy in guiding the search toward higher-quality solutions. In particular, for more complex instances, the learning mechanism enables the algorithm to select more suitable destroy-and-repair operators dynamically, thereby improving neighborhood exploration efficiency, search robustness, and overall optimization performance.

Table 10
Effectiveness analysis of the Q-learning-based operator selection mechanism.

Instance	QL-IPSO-ALNS			RW-IPSO-ALNS				
	Opt	Ave	Time(s)	Opt	Ave	Time(s)	Gap1(%)	Gap2(%)
C101_30	720.43	730.00	2.48	720.43	737.09	2.71	0.00%	0.96%
C101_50	1295.00	1300.95	19.69	1288.55	1307.84	17.66	−0.50%	0.53%
C101_75	2320.82	2339.73	72.91	2334.25	2357.88	67.42	0.58%	0.77%
C101_100	2976.48	3014.66	355.21	3003.26	3049.99	224.20	0.89%	1.16%
R101_30	1367.81	1393.93	1.22	1367.81	1400.24	1.53	0.00%	0.45%
R101_50	1900.01	1933.59	8.57	1927.60	1956.27	8.09	1.43%	1.16%
R101_75	2695.38	2758.33	51.53	2710.20	2789.77	48.25	0.55%	1.13%
R101_100	3221.90	3334.57	262.48	3234.67	3376.70	198.0	0.39%	1.25%
RC101_30	1488.19	1488.19	2.92	1488.19	1491.01	2.41	0.00%	0.19%
RC101_50	1879.34	1885.11	21.21	1879.34	1884.33	17.20	0.00%	−0.04%
RC101_75	2994.24	3040.43	61.67	3017.34	3076.00	54.87	0.77%	1.16%
RC101_100	3563.60	3682.66	187.14	3639.40	3753.68	142.99	2.08%	1.89%
Average	–	–	–	–	–	–	0.52%	0.88%

5.5. Sensitivity analysis

To further examine the performance of the proposed model and algorithm, a sensitivity analysis is conducted from four perspectives. First, the pure truck mode is compared with collaborative modes in which each truck carries one or two drones. Second, the effect of urgency-aware delivery deadlines on routing decisions is investigated. Third, the influence of drone endurance is examined. Fourth, the sensitivity of system performance to variations in the cost weights in the objective function is analyzed.

5.5.1. Fleet configuration analysis

To quantify the effect of drone fleet size on the performance of the collaborative delivery system, experiments are conducted on three representative Solomon instance types: C-type (60 affected sites), R-type (38 affected sites), and RC-type (30 affected sites). Three delivery configurations are considered: truck-only delivery, truck with one drone, and truck with two drones. With all other parameters fixed, total system cost and completion time are compared across the three configurations. The results are shown in Fig. 7.

As illustrated in Fig.7, both collaborative delivery modes consistently outperform the truck-only mode across all three instance types, which indicates that the adoption of drones enhances the economic efficiency of emergency relief distribution. A comparison of two collaborative configurations reveals that the two-drone mode generally achieves superior performance to the one-drone mode; however, the degree of improvement exhibits substantial variability depending on the spatial distribution of demand. For C-type instances, the total cost is reduced by approximately 3.02% with one drone per truck and 9.98% with two drones per truck relative to the truck-only mode. For R-type instances, the corresponding reductions are 3.79% and 7.88%, whereas for RC-type instances they are 2.54% and 2.81%, respectively. These results indicate that adding drones can further reduce system cost, but the marginal benefit is not uniform and is particularly limited for RC-type instances.

A similar pattern is observed for completion time. Compared with the truck-only model, both collaborative modes reduce total route completion time across all three instance classes, and increasing the number of drones yields further reductions. For C-type instances, the reductions are approximately 1.24% and 2.96% for the one-drone and two-drone configurations, respectively. For R-type instances, the corresponding reductions are 1.84% and 4.06%, and for RC-type instances they are 1.89% and 2.52%. Overall, drones improve both economic performance and time performance, but the extent of improvement is highly dependent on the spatial distribution of affected sites and the proportion of sites that can be feasibly served by drones. Therefore, increasing the number of drones does not necessarily guarantee better performance, and the fleet configuration should be matched to the characteristics of the service region.

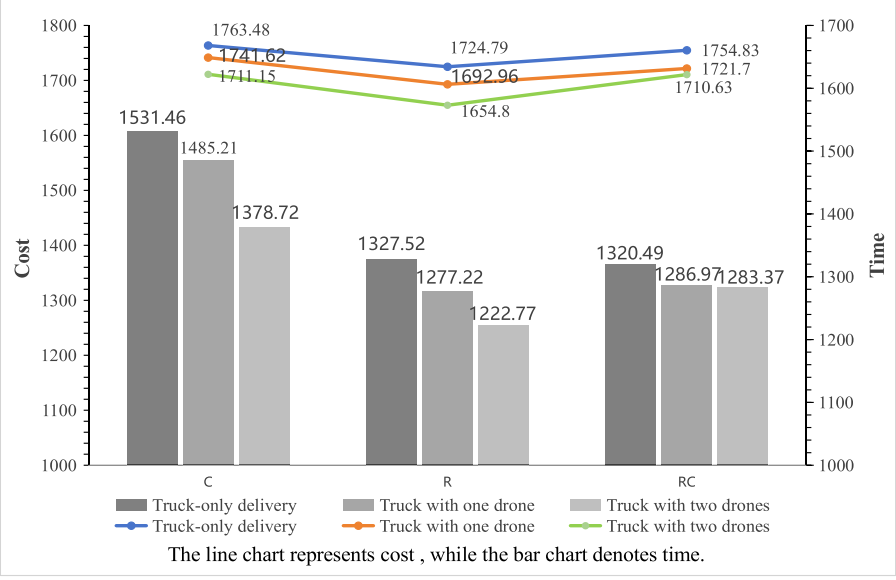


Fig. 7. Cost comparison across different grouping modes.

5.5.2. Effect of urgency-aware delivery deadlines on routing decisions

To further examine how urgency-aware delivery deadlines affect operational decisions, a comparative analysis is conducted under the truck–drone collaborative delivery mode for scenarios with and without urgency-dependent deadline adjustments. Five independent runs are performed, and a representative solution is selected from each run for illustration, as shown in Fig. 8.

The results show that increasing the urgency level of an affected site progressively tightens its effective delivery deadline, thereby changing both service sequencing and vehicle assignment. High-priority affected sites tend to be served earlier, while drone delivery is preferentially assigned when a feasible drone sortie can satisfy the corresponding deadline. Consequently, the routing structure may deviate from the shortest-distance solution in order to accommodate urgent delivery requirements.For example, after serving affected site 26, Truck 3 would normally proceed directly toward site 28 because this route is more efficient in terms of travel distance. However, sites 25 and 30 have higher rescue priorities and therefore tighter effective delivery deadlines. To satisfy these requirements, Truck 3 adjusts its route and serves the urgent sites earlier, even at the expense of additional travel distance. Meanwhile, drones are more frequently assigned to remote or time-sensitive affected sites, allowing urgent deliveries to be completed without substantially disrupting the main truck routes.

These results reveal an important trade-off between routing efficiency and emergency responsiveness. Tighter urgency-aware deadlines restrict the feasible solution space and may slightly increase total system completion time or transportation cost. However, they substantially improve the timeliness of service for high-priority affected sites. More importantly, the results demonstrate that the proposed deadline-adjustment mechanism can translate field-assessed rescue priorities into explicit routing and service decisions, rather than treating urgency as an external qualitative consideration. This provides a more operationally meaningful representation of priority differentiation in post-disaster relief distribution.

5.5.3. Effect of drone endurance

A sensitivity analysis is conducted on the C-, R-, and RC-type instances by varying drone endurance from 20 min to 40 min. The resulting changes in total cost and total route completion time are shown in Fig. 9.

Overall, cost and total route completion time respond differently to endurance improvements across the three instance types. In terms of cost, R-type and RC-type instances are more sensitive to changes in endurance. For R-type instances, total cost decreases from approximately 1861.20 to 1738.60, corresponding to a reduction of about 6.6%. For RC-type instances, total cost decreases from 1668.78 to 1596.16, corresponding to a reduction of about 4.4%. By contrast, C-type instances show a cost reduction of $<1\%$, indicating that even a relatively low battery level is sufficient to complete most candidate drone tasks, and therefore increasing the drone endurance does not significantly reduce the total cost.

In terms of time performance, RC-type instances exhibit the strongest response. Total route completion time decreases from 1219.90 to 1114.42, a reduction of about 8.6%. For R-type instances, total route completion time decreases from 1426.50 to 1400.00, corresponding to a reduction of about 1.9%, while for C-type instances it decreases from 1416.71 to 1385.76, corresponding to a reduction of about 2.2%.

These results confirm that increasing drone endurance expands the feasible service region of drones and allows more tasks to be shifted away from trucks to drones. However, the magnitude of the benefit depends strongly on affected site distribution. In clustered scenarios, the marginal gain is limited because the original endurance is already adequate. In mixed scenarios, by contrast, endurance enhancement improves both service coverage and task parallelism, leading to more substantial system-level gains.

5.5.4. Effect of objective weight coefficients

To examine the trade-off between economic cost and delivery timeliness, a sensitivity analysis is conducted on the distance-cost weight α and the time-cost weight β in the objective function. Using C-type, R-type, and RC-type Solomon instances, the ratio $\alpha : \beta$ is varied from 3:7 to 7:3, and the resulting distance cost and total delivery time are recorded. The results are shown in Fig. 10.

For C-type instances, increasing α from 0.3 to 0.7 reduces distance cost from 3384.20 to 3277.26, corresponding to a reduction of approximately 3.2%, while total delivery time increases from 2392.88 to 2449.81, corresponding to an increase of approximately 2.4%. These changes are relatively moderate, indicating that the trade-off between distance cost and delivery time is less pronounced in clustered scenarios.

For R-type instances, increasing the distance weight yields a more pronounced reduction in distance cost. Specifically, distance cost decreases from 3856.87 to 3695.34, a reduction of 4.2%, whereas total delivery time increases from 2629.37 to 2680.55, an increase of approximately 1.9%. Compared with the C-type instances, the random instances offer greater potential for reducing cost, but this comes at the expense of response speed.

For RC-type instances, the response is more complex. As α increases, distance cost decreases from 4339.82 to 4129.16, corresponding to a cumulative reduction of approximately 4.9%. However, total delivery time changes non-monotonically, first increasing from 2898.89 to

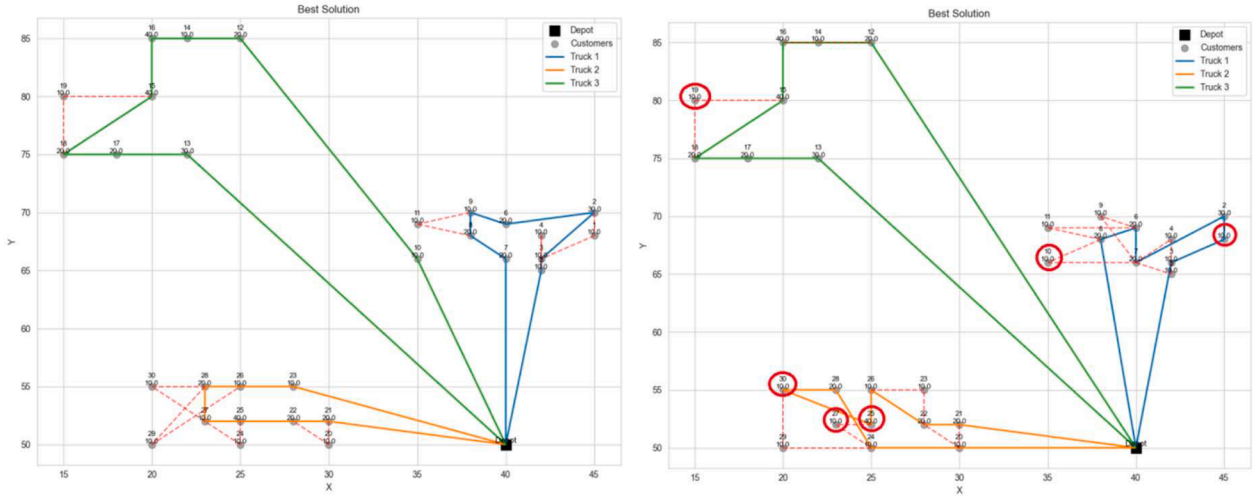


Fig. 8. Route comparison with and without urgency-aware delivery deadlines.

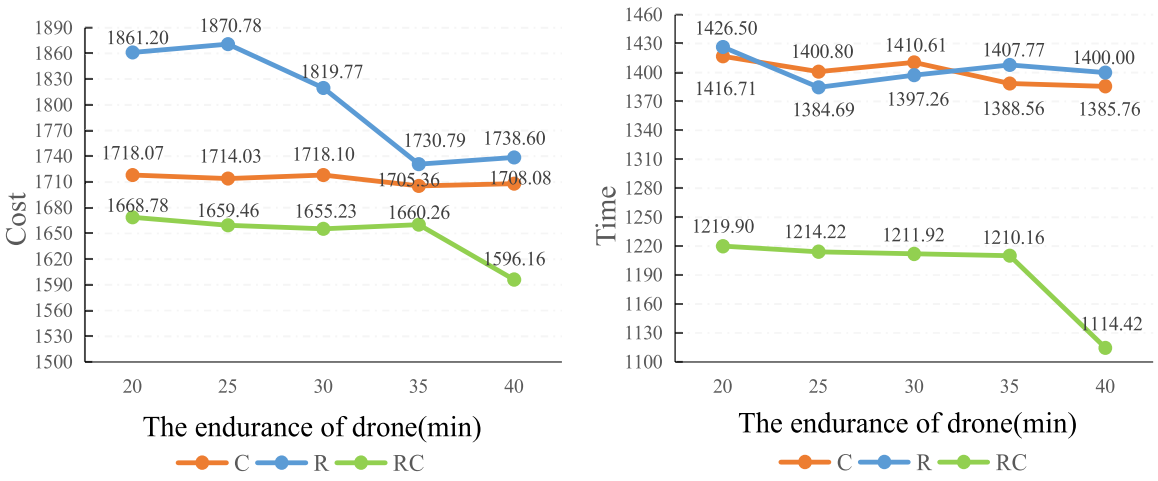


Fig. 9. Cost variation of drones with different endurance levels.

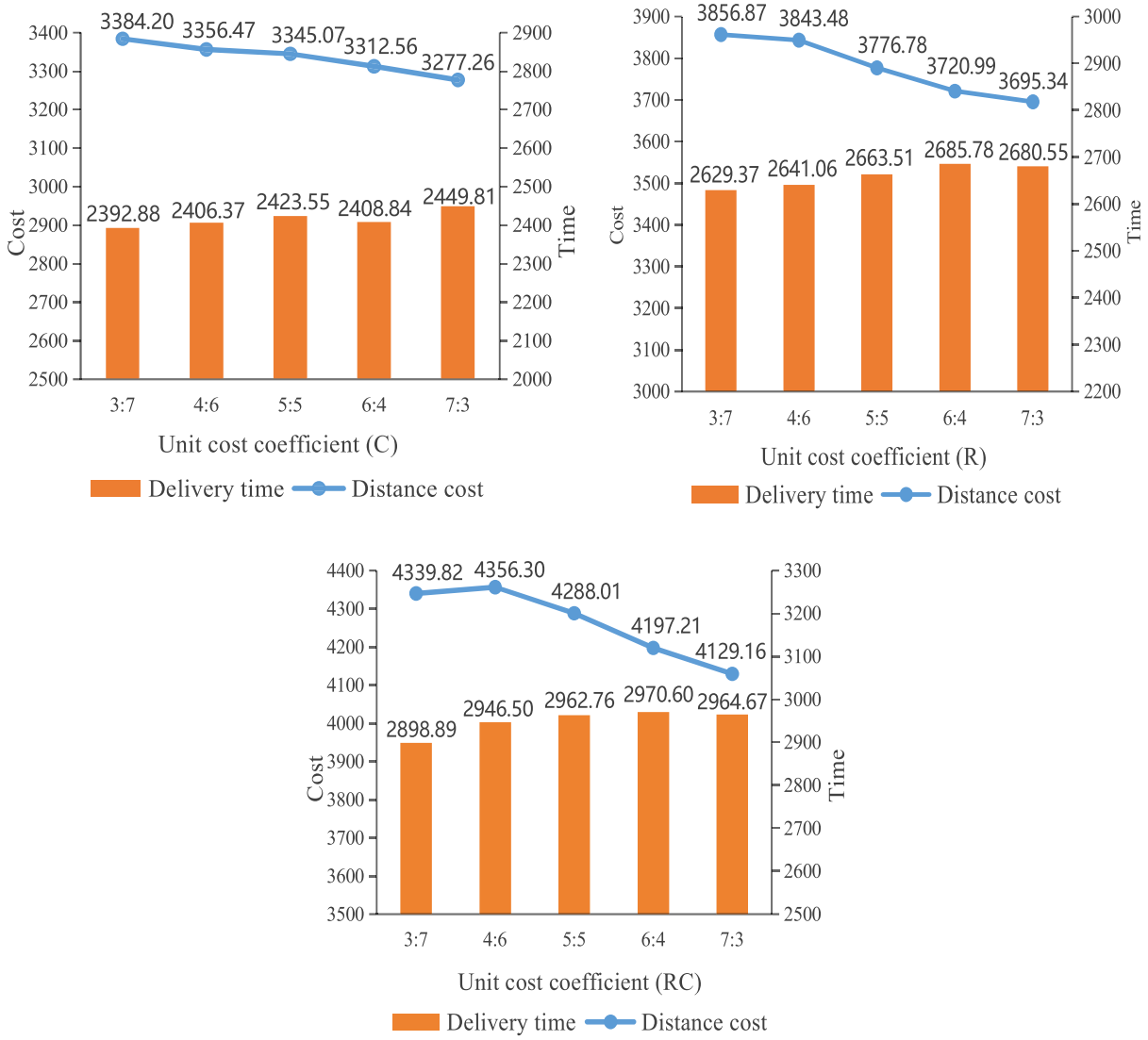


Fig. 10. Sensitivity to unit cost coefficient variations in the objective function.

2970.60 and then slightly decreasing to 2964.67, although it remains on an overall upward trend. This suggests that, in mixed-distribution scenarios, the solution structure may change discretely as the objective weights vary.

In summary, increasing the distance-cost weight α consistently reduces distance cost across all three instance classes, whereas total delivery time generally increases, reflecting the expected trade-off between economic efficiency and response speed. The results indicate that decision-makers can reasonably adjust the weights between relief cost and efficiency according to different post-disaster response stages.

6. Case analysis

To further evaluate the practical applicability of the proposed VRPMTMD model and QL-IPSO-ALNS algorithm, a real-world case study is conducted based on the July 20, 2021 heavy rainfall disaster in Zhengzhou, Henan Province, China. The event caused severe casualties and substantial economic losses and highlighted the challenges of emergency material allocation and rapid rescue response under disrupted urban conditions.

Following Wu [50], the case study focuses on Jinshui District of Zhengzhou. The emergency material reserve warehouse in the district is designated as the distribution depot, while 40 affected sites are considered demand nodes. For each affected site, geographical coordinates, material demand, and rescue urgency are specified. The resulting dataset provides a realistic spatial and demand structure for evaluating truck–drone collaborative emergency distribution. The detailed information is reported in Table 11 and the spatial distribution of the depot and affected sites is illustrated in Fig. 11.

To maintain consistency with the numerical experiments in Section 5, the fleet consists of four trucks, each equipped with two drones. The truck and drone capacities, endurance, travel speeds, and cost parameters are set according to the experimental configuration described in Section 5.1. For both transportation modes, the distance matrix is constructed from the Euclidean distances between node coordinates. For trucks, the Euclidean distance is used as a simplified approximation of road travel distance, whereas for drones it represents the corresponding flight distance.

The resulting collaborative delivery plan partitions the 40 affected sites among four truck-based service regions. Among these sites, 30are served directly by trucks, while 10 are served by drones through 10 drone missions. Truck 3 performs six drone missions, whereas Trucks 1, 2, and 4 each perform two, one, and one mission, respectively. The drone-served sites are mainly associated with high rescue urgency or relatively unfavorable spatial locations, indicating that the proposed model can effectively exploit drones to provide rapid service to time-sensitive and spatially dispersed demand points.

A representative example occurs at site 24, where both drones carried by Truck 3 are launched simultaneously. Drone 1 performs mission (24, 13, 12), while Drone 2 performs mission (24, 35, 12), with both drones recovering at site 12. This parallel deployment demonstrates the ability of the proposed model to coordinate multiple drones from the same launch node and serve multiple urgent sites simultaneously, thereby improving delivery flexibility and reducing potential response delays (Table 12).

The urgency-aware time-window mechanism also influences routing decisions. Sites with higher urgency levels, such as sites 18 and 35, are assigned tighter delivery deadlines and are preferentially served by drones to ensure timely response. By contrast, sites with lower urgency or larger demand quantities are typically served directly by trucks to better utilize vehicle load capacity. This allocation reflects the trade-off between urgency satisfaction and operational efficiency embedded in the proposed model.

The case results demonstrate that the proposed model can generate a practical truck-drone relief distribution plan for real post-disaster scenarios, balancing transportation cost, delivery time, and urgency-aware constraints. The approach provides decision support for emergency logistics managers to improve the efficiency and responsiveness of relief operations (Fig. 12).

7. Managerial implications

Based on the computational experiments, sensitivity analyses, and case study, several managerial implications can be derived for emergency logistics planning and truck–drone fleet deployment.

(1) Delivery deadlines should be differentiated according to the urgency of affected sites. The sensitivity analysis shows that urgency-aware delivery deadlines can influence both service-mode selection and routing decisions. More urgent affected sites tend to receive tighter delivery deadlines and are more likely to be assigned to drones, allowing limited aerial resources to be directed toward time-critical tasks. Therefore, emergency logistics managers should avoid applying identical delivery requirements to all affected sites. Instead, urgency levels should be incorporated into deadline setting and continuously considered

Table 11
Coordinates, demand quantities, and urgency levels of demand nodes.

Number	Longitude	Latitude	Demand	Priority	Number	Longitude	Latitude	Demand	Priority
0	113.677384	34.797821	0	0	21	113.807973	34.805686	5	4
1	113.650885	34.801918	9	1	22	113.787106	34.7891	11	1
2	113.662505	34.782285	8	3	23	113.820079	34.783654	6	2
3	113.670116	34.766761	10	2	24	113.757597	34.786621	8	1
4	113.674632	34.813708	11	1	25	113.783541	34.767899	8	1
5	113.664981	34.839578	4	1	26	113.818075	34.843075	10	3
6	113.688942	34.841079	7	1	27	113.825062	34.871898	5	1
7	113.730631	34.849	12	2	28	113.783549	34.832427	7	1
8	113.715974	34.833186	8	1	29	113.794259	34.781315	12	2
9	113.700412	34.803239	10	1	30	113.680603	34.778635	21	1
10	113.718989	34.757686	8	1	31	113.692701	34.762581	6	1
11	113.702409	34.752795	8	1	32	113.70541	34.782166	6	1
12	113.734408	34.786355	6	2	33	113.664417	34.827453	10	1
13	113.742743	34.796075	7	1	34	113.819906	34.811408	11	1
14	113.746311	34.830111	4	1	35	113.748963	34.790417	6	5
15	113.762564	34.819848	10	3	36	113.767274	34.778336	8	1
16	113.76367	34.841771	6	1	37	113.737975	34.752771	16	1
17	113.785058	34.821135	6	1	38	113.746288	34.766884	11	1
18	113.785058	34.804437	5	5	39	113.811759	34.767857	16	1
19	113.802404	34.835126	5	1	40	113.713059	34.790895	4	2
20	113.806269	34.813428	11	1	–	–	–	–	–

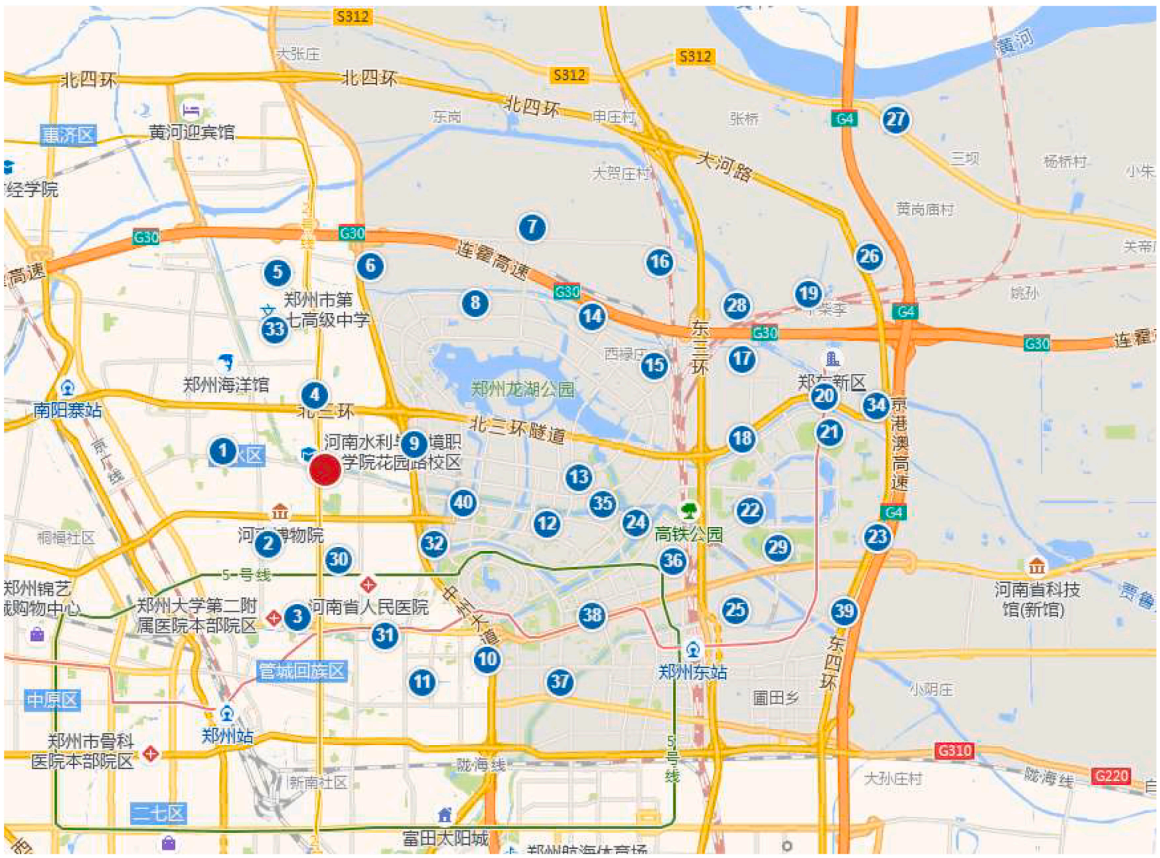


Fig. 11. Spatial distribution of demand nodes and depot.

Table 12
Truck-drone collaborative delivery scheme.

Sub-routes	Delivery vehicles	Specific route
1	Truck	0→32→40→38→37→10→11→3→30→0
	Drone1	11→31→30
	Drone2	3→2→30
2	Truck	0→4→6→33→1→0
	Drone1	6→5→33
	Drone2	– – –
3	Truck	0→15→17→20→34→23→39→25→22→24→12→0
	Drone1	17→18→22, 22→36→24, 24→13→12
	Drone2	20→21→34, 25→29→22, 24→35→12
4	Truck	0→8→7→16→27→26→28→14→9→0
	Drone1	26→19→28
	Drone2	– – –

when assigning truck and drone resources. This enables limited emergency resources to be prioritized for the most critical locations while maintaining overall delivery efficiency.

(2) Truck–drone fleet configuration should be determined according to operational requirements rather than simply increasing the number of drones. The results indicate that drones can improve delivery efficiency by providing rapid and flexible service, but the benefits depend on the overall fleet configuration and operational environment. Therefore, emergency planners should determine the number of trucks and drones jointly according to the number of affected sites, demand characteristics, and response requirements. A balanced truck–drone configuration can avoid excessive investment in aerial resources while maintaining sufficient delivery capacity and operational flexibility.

(3) Drone endurance is an important operational parameter for fleet planning. Increasing drone endurance expands the feasible service range and allows more affected sites to be transferred from truck service to drone service. However, the magnitude of the benefit varies across different spatial distributions, with greater improvements observed in more dispersed or mixed-demand scenarios. Therefore, investment in higher-endurance drones should be evaluated together with the geographic distribution of demand. When affected sites are already concentrated within a relatively short flight range, additional endurance may provide limited operational benefits.

(4) The relative importance of cost and delivery time should be adjusted according to the emergency response stage. The sensitivity analysis demonstrates a clear trade-off between transportation cost and delivery time. Increasing the weight assigned to transportation cost can reduce distance-related expenditure but may increase total delivery time. In the early response stage, when rapid relief delivery is the primary concern, decision-makers may place greater emphasis on delivery time. As operations gradually shift toward replenishment and sustained distribution, greater attention can be given to transportation efficiency and operating cost. Thus, the weighting of economic and time-related objectives should reflect the priorities of different emergency response stages.

8. Conclusions

This study investigates the vehicle routing problem with multiple trucks and drones for emergency relief distribution. A mixed-integer programming model is formulated by considering urgency-aware delivery deadlines and service times, with the objective of minimizing the weighted transportation cost and delivery time costs. To solve this NP-

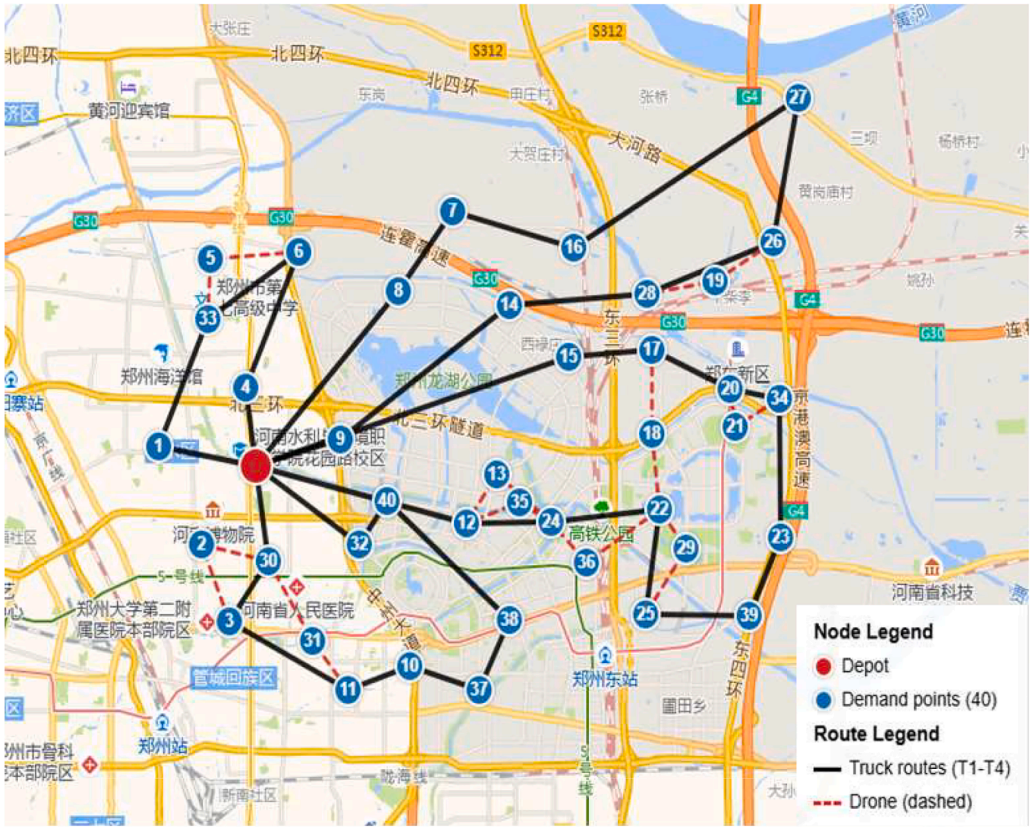


Fig. 12. Route visualization of truck-drone collaborative delivery.

hard problem efficiently, a QL-IPSO-ALNS algorithm is proposed, which incorporates the global search capability of IPSO, the local refinement ability of ALNS and an adaptive operator selection mechanism using Q-learning to improve solution quality and computational efficiency. Numerical experiments on Solomon-derived instances show that QL-IPSO-ALNS obtains near-optimal solutions for small- and medium-scale cases, with an average deviation from the Gurobi optimum below 0.55%, while greatly reducing computation time. For large-scale instances, QL-IPSO-ALNS outperforms IPSO, K-means-ALNS, LNS, CH-SaBO, and MAAGLNS in terms of solution quality, stability, and computational efficiency. The urgency-aware deadline adjustment is further shown to steer delivery resources toward severely affected sites at a modest increase in total cost, and a real-world case study demonstrates the practical applicability of the proposed approach. These results demonstrate the effectiveness and scalability of the proposed approach for large-scale emergency truck-drone collaborative routing.

Despite these contributions, several limitations remain. The current study is developed within a relatively static emergency scheduling framework and does not fully capture dynamic changes in demand, road conditions, or task priorities that may arise in real disaster environments. In addition, to maintain model tractability, some practical factors related to drone operations, such as battery limitations, weather impacts, and more complex synchronization constraints, are simplified. Future research may therefore extend the proposed framework to dynamic and stochastic emergency scenarios, incorporate more realistic operational constraints, and further improve the solution approach through adaptive learning mechanisms or more efficient computational strategies. These extensions would further enhance the applicability of the proposed model and algorithm in large-scale and complex emergency logistics settings.

CRediT authorship contribution statement

Xiaofei Kou: Supervision, Project administration, Methodology, Funding acquisition, Conceptualization, Writing – original draft, Software. **Tianlong Liu:** Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Hao Liu:** Writing – review & editing, Validation, Resources, Investigation, Data curation. **Gangyan Xu:** Writing – review & editing, Validation, Resources, Investigation, Formal analysis. **Peng Sun:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by the Humanities and Social Sciences Project of the Ministry of Education (no. 24YJC630095), the Science and Technology Research Project of Henan Province (nos. 262102320070, 262102240015, 252102320128), the Key Scientific Research Project of Henan Provincial Department of Transport(No. 2026-15),the National Natural Science Foundation of China (nos. 72501098, 72101178, 52575603) and the Graduate Education Innovation Program Fund of Zhengzhou University of Aeronautics (no. 2025CX38).

Data availability

Data will be made available on request.

References

[1] X. Zhao, H. Li, Q. Cai, Y. Pan, Y. Qi, Managing extreme rainfall and flooding events: a case study of the 20 July 2021 Zhengzhou flood in China, Climate 11 (11) (2023) 228, https://doi.org/10.3390/cli11110228.

[2] C. Xie, C. Xu, Y. Huang, J. Liu, X. Shao, X. Xu, Z. Xiao, Advances in the study of natural disasters induced by the "23.7" extreme rainfall event in North China, Nat. Hazards Res. 5 (1) (2025) 1–13.

[3] L. Johnson, 9 things you need to know about the Amazon Prime Air drone delivery service, Digit. Spy (2017). Retrieved from, https://www.digitalspy.com/tech/feature/a820748/amazon-prime-air-drone-delivery-service/. Accessed November 18, 2025.

[4] C. Shu, Flytrex Launches an Autonomous On-Demand Drone Delivery Service in Iceland's Capital, 2017. Retrieved from, http://techcrunch.com/2017/08/23/flytrex-launches-an-autonomous-on-demand-drone-delivery-service-in-icelands-capital/. Accessed July 20, 2018.

[5] J. Porter, Google's wing drones approved to make public deliveries in Australia, The Verge (2019). Retrieved from, https://www.theverge.com/2019/4/9/18301782/wing-drone-delivery-google-alphabet-canberra-australia-public-launch. Accessed June 21, 2019.

[6] C.C. Murray, A.G. Chu, The flying sidekick traveling salesman problem: optimization of drone-assisted parcel delivery, Transp. Res. C: Emerg. Technol. 54 (2015) 86–109, https://doi.org/10.1016/j.trc.2015.03.005.

[7] Q.M. Ha, Y. Deville, Q.D. Pham, M.H. Hà, On the min-cost traveling salesman problem with drone, Transp. Res. C: Emerg. Technol. 86 (2018) 597–621, https://doi.org/10.1016/j.trc.2017.11.015.

[8] N. Agatz, P. Bouman, M. Schmidt, Optimization approaches for the traveling salesman problem with drone, Transp. Sci. 52 (4) (2018) 965–981, https://doi.org/10.1287/trsc.2017.0791.

[9] D. Etherington, Check Out the Real World Version of the Mercedes-Benz Drone Delivery Van, 2017. Retrieved from, https://techcrunch.com/2017/01/10/check-out-the-real-world-version-of-the-mercedes-benz-drone-delivery-van/. Accessed March 10, 2025.

[10] D. Buchmueller, S.A. Green, A. Kalyan, G. Kimchi, Communications and Landings of Unmanned Aerial Vehicles on Transportation Vehicles for Transport, Google Patents, 2020.

[11] P. Kitjacharoenchai, B.C. Min, S. Lee, Two echelon vehicle routing problem with drones in last mile delivery, Int. J. Prod. Econ. 225 (2020) 107598.

[12] J.J. Yoon, The Traveling Salesman Problem with Multiple Drones: An Optimization Model for Last-Mile Delivery, Massachusetts Institute of Technology, 2018 (Doctoral dissertation).

[13] C.C. Murray, R. Raj, The multiple flying sidekicks traveling salesman problem: parcel delivery with multiple drones, Transp. Res. C: Emerg. Technol. 110 (2020) 368–398.

[14] S. Poikonen, B. Golden, Multi-visit drone routing problem, Comput. Oper. Res. 113 (2020) 104802.

[15] X. Wang, S. Poikonen, B. Golden, The vehicle routing problem with drones: several worst-case results, Optim. Lett. 11 (4) (2017) 679–697.

[16] S. Poikonen, X. Wang, B. Golden, The vehicle routing problem with drones: extended models and connections, Networks 70 (1) (2017) 34–43.

[17] D. Schermer, M. Moeini, O. Wendt, A matheuristic for the vehicle routing problem with drones and its variants, Transp. Res. C: Emerg. Technol. 106 (2019) 166–204.

[18] P. Kitjacharoenchai, M. Ventresca, M. Moshref-Javadi, S. Lee, J.M. Tanchoco, P. A. Brunese, Multiple traveling salesman problem with drones: mathematical model and heuristic approach, Comput. Ind. Eng. 129 (2019) 14–30.

[19] M. Dell'Amico, R. Montemanni, S. Novellani, Algorithms based on branch and bound for the flying sidekick traveling salesman problem, Omega-Int. J. Manag. Sci. 104 (2021) 102493, https://doi.org/10.1016/j.omega.2021.102493.

[20] S. Poikonen, B. Golden, E.A. Wasil, A branch-and-bound approach to the traveling salesman problem with a drone, INFORMS J. Comput. 31 (2) (2019) 335–346, https://doi.org/10.1287/ijoc.2018.0826.

[21] F. Tamke, U. Buscher, A branch-and-cut algorithm for the vehicle routing problem with drones, Transp. Res. B: Methodol. 144 (2021) 174–203, https://doi.org/10.1016/j.trb.2020.11.011.

[22] K. Wang, B. Yuan, M. Zhao, Y. Lu, Cooperative route planning for the drone and truck in delivery services: a bi-objective optimisation approach, J. Oper. Res. Soc. 71 (10) (2020) 1657–1674, https://doi.org/10.1080/01605682.2019.1621671.

[23] S. Huang, Y. Huang, C.A. Blazquez, C. Chen, Solving the vehicle routing problem with drone for delivery services using an ant colony optimization algorithm, Adv. Eng. Inform. 51 (2022) 101536, https://doi.org/10.1016/j.aei.2022.101536.

[24] B. Lan, Y. Suzuki, Traveling salesman problem with time windows and a drone-utilizing intermediate points (TSPTWD-IP), Comput. Ind. Eng. 198 (2024) 110641.

[25] Y. Luo, X. Deng, W. Zhang, Y. Ke, S. Wan, Y. Qian, Collaborative intelligent delivery with one truck and multiple heterogeneous drones in COVID-19 pandemic environment, IEEE Trans. Intell. Transp. Syst. 25 (2024) 7907–7920, https://doi.org/10.1109/TITS.2024.3350876.

[26] J. Xiao, Y. Li, Z. Cao, J. Xiao, Cooperative trucks and drones for rural last-mile delivery with steep roads, Comput. Ind. Eng. 187 (2024) 109849, https://doi.org/10.1016/j.cie.2023.109849.

[27] T. Guo, Y. Mei, M. Zhang, H. Zhao, K. Cai, W. Du, Learning-aided neighborhood search for vehicle routing problems, IEEE Trans. Pattern Anal. Mach. Intell. 47 (7) (2025) 5930–5944.

[28] Y. Shi, K. Zhang, J. Zhang, M. Chen, J. Yang, H. Guo, Dynamic truck-drone cooperative delivery of emergency supplies considering secondary disasters, Transp. Res. E: Logist. Transp. Rev. 206 (2026) 104543.

[29] Q. Guo, Y. Qiu, Y. Shi, Q. Zheng, H. Guo, An enhanced dual-population NSGA-II for solving multi-objective emergency supplies dispatch problem based on truck-drone-dispatch vehicle collaboration, Comput. Oper. Res. 187 (2026) 107324.

[30] M.A. Ertem, M. İşbilir, A. Şahin Arslan, Review of intermodal freight transportation in humanitarian logistics, Eur. Transp. Res. Rev. 9 (1) (2017) 10.

[31] N. Al Theeb, C. Murray, Vehicle routing and resource distribution in post-disaster humanitarian relief operations, Int. Trans. Oper. Res. 24 (6) (2017) 1253–1284.

[32] Q. Shi, W. Yang, Z.M. Yang, Q.C. Zhao, Relief materials vehicles planning in natural disasters, IEEE/CAA J. Autom. Sin. 5 (2) (2018) 595–601.

[33] Y. Zhang, J. Liu, Emergency logistics scheduling under uncertain transportation time using online optimization methods, IEEE Access 9 (2021) 36995–37010.

[34] Y. Lu, C. Yang, J. Yang, A multi-objective humanitarian pickup and delivery vehicle routing problem with drones, Ann. Oper. Res. 319 (1) (2022) 291–353.

[35] L. Zhang, Y. Ding, H. Lin, Optimizing synchronized truck-drone delivery with priority in disaster relief, J. Ind. Manag. Optim. 19 (7) (2023) 5143–5162.

[36] Y. Long, G. Xu, J. Zhao, B. Xie, M. Fang, Dynamic truck–UAV collaboration and integrated route planning for resilient urban emergency response, IEEE Trans. Eng. Manag. 71 (2024) 9826–9838.

[37] X. Weng, W. She, H. Fan, J. Zhang, L. Yun, Multi-depot vehicle routing problem with drones in emergency logistics, Clust. Comput. 28 (1) (2025) 64.

[38] Y. Shi, J. Yang, Q. Han, H. Song, H. Guo, Optimal decision-making of post-disaster emergency material scheduling based on helicopter–truck–drone collaboration, Omega 127 (2024) 103104.

[39] T. Guo, Y. Mei, M. Zhang, H. Ding, W. Du, Multitree genetic programming with behavioral semantics for large-scale dynamic air traffic flow management, IEEE Trans. Evol. Comput. (2026).

[40] Y. Yin, Y. Yang, Y. Yu, D. Wang, T.C.E. Cheng, Robust vehicle routing with drones under uncertain demands and truck travel times in humanitarian logistics, Transp. Res. B: Methodol. 174 (2023) 102781.

[41] R. Van Steenbergen, M. Mes, W. Van Heeswijk, Reinforcement learning for humanitarian relief distribution with trucks and UAVs under travel time uncertainty, Transp. Res. C: Emerg. Technol. 157 (2023) 104401.

[42] J. Sun, G.H. Lin, X. Zhu, A hybrid drone-truck model for fair and efficient emergency relief distribution, Comput. Ind. Eng. 211 (2026) 111607.

[43] J. Lu, Y. Liu, C. Jiang, W. Wu, Truck-drone joint delivery network for rural area: optimization and implications, Transp. Policy 163 (2025) 273–284.

[44] J. Kennedy, R. Eberhart, Particle swarm optimization, in: Proceedings of ICNN'95-International Conference on Neural Networks 4, ieee, 1995, pp. 1942–1948.

[45] S. Li, M. Tan, I.W. Tsang, J.T.-Y. Kwok, A hybrid PSO-BFGS strategy for global optimization of multimodal functions, IEEE Trans. Syst. Man Cybern. B (Cybern.) 41 (4) (2011) 1003–1014.

[46] L. Tan, A clustering K-means algorithm based on improved PSO algorithm, in: 2015 Fifth International Conference on Communication Systems and Network Technologies, IEEE, 2015, pp. 940–944.

[47] S. Ropke, D. Pisinger, An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows, Transp. Sci. 40 (4) (2006) 455–472.

[48] R. Zhai, X. Zhang, Y. Mei, T. Guo, W. Du, Co-evolution with hierarchical decomposition for vehicle routing problem with drones, Swarm Evol. Comput. 105 (2026) 102402.

[49] Y. Chen, F. Hong, Q. Ye, G. Wu, Path planning for large-area search by a collaborative truck-UAVs system, Expert Syst. Appl. 331 (2026) 133385.

[50] J. Wu, Truck–Drone Collaborative Delivery Route Optimization Method for Emergency Supplies in Urban Waterlogging Scenarios (Master's Thesis), Southeast University, Nanjing, China, 2025.